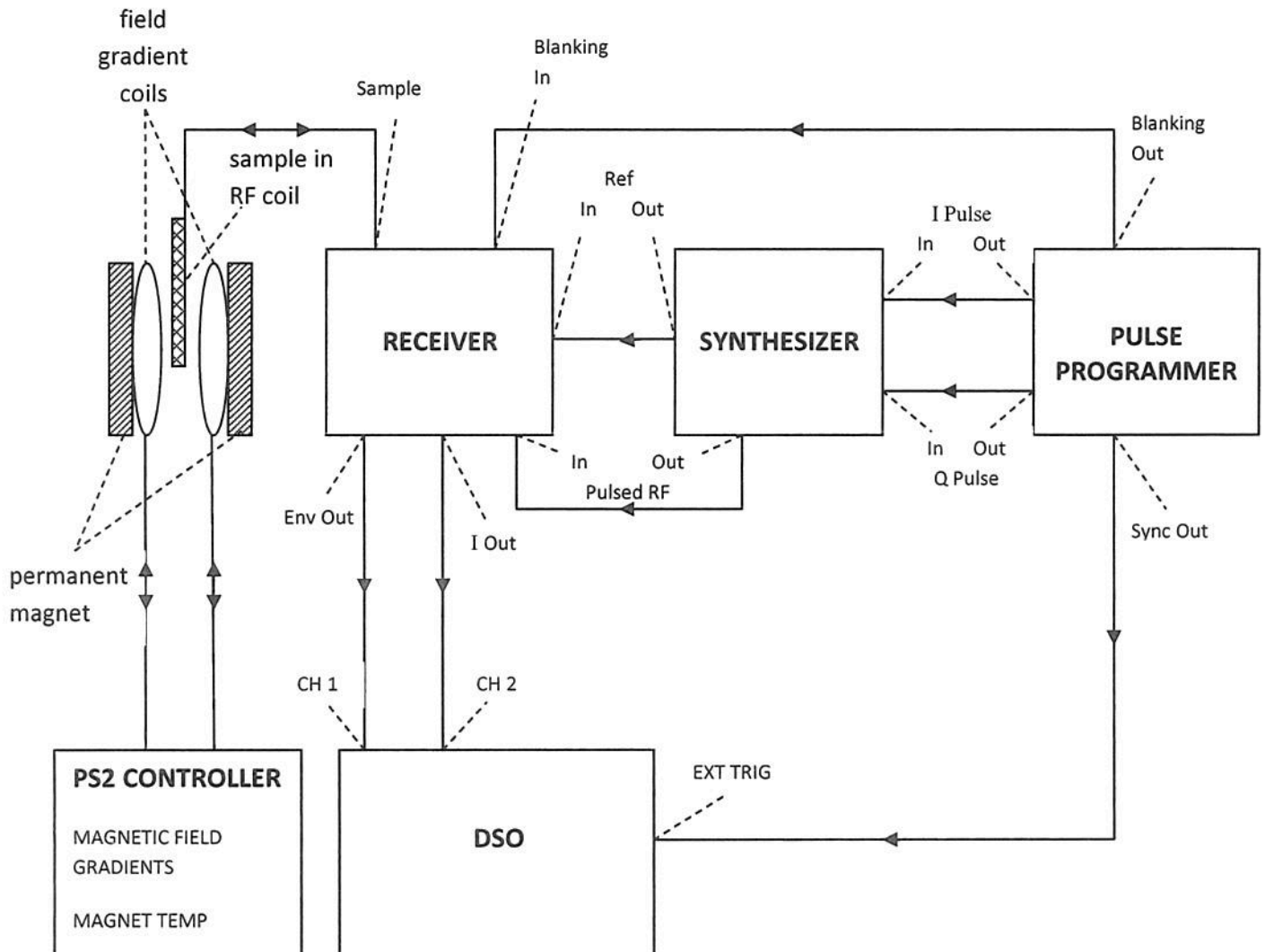


PULSED NUCLEAR MAGNETIC RESONANCE

Spectrometer Connection and Initial Set-up



PS2 Spectrometer connection diagram

Before switching anything on:

Find the light oil (L– OIL) sample and adjust the rubber O-ring on the sample tube so that the ring's **bottom** is 43 mm above the **outside bottom** of the sample tube.

Insert the sample tube into the hole in the centre of the PERMANENT MAGNET.

DO NOT FORCE IT IN, UNDER ANY CIRCUMSTANCES! – if it doesn't drop in easily, get a demonstrator.

The rest of the set-up is carried out with the L-OIL sample correctly positioned in the sample coil.

PS2 CONTROLLER (stabilises the temperature of the permanent magnet and controls the currents through the field gradient coils)

RIGHT – FEEDBACK LOOP – OPEN

LEFT - FEEDBACK LOOP – OPEN

MAGNETIC FIELD GRADIENTS

X: + 1.00 Y: - 5.50 Z: - 0.30 Z2: - 3.50

21 MHz Receiver (receives and amplifies the signal from the NMR, RF coil)

Pulsed RF In (cable from Synthesizer, Pulsed RF Out)

Sample: (blue cable from **PERMANENT MAGNET**)

Gain: .75 Band: p

Blanking: On Width: 12 o'clock

In (cable from **Programmer**, Blanking Out)

Filter TC: .01 ms

Ref In (cable from **Synthesizer**, Ref Out)

I Out (cable to **DSO**, CH 2)

Env. Out (cable to **DSO**, CH 1)

21 MHz Synthesizer (generates the RF coil excitation current)

Ref Out: On (cable to **Receiver**, Ref In)

CW Out: Off

I Pulse In (cable from **Programmer**, I Pulse Out)

Q Pulse In (cable from **Programmer**, Q Pulse Out)

Pulsed RF Out (cable to **Receiver**, Pulsed RF In)

Pulse Programmer (generates two or more pulses with adjustable length and separation)

Blanking Out (to **Receiver**, Blanking In)

MG: Off

Sync: A

Sync Out (cable to **DSO**, EXT TRIG)

Pulse A: On Pulse B: Off

I Pulse Out (cable to **Synthesizer**, I Pulse In)

Q Pulse Out (cable to **Synthesizer**, Q Pulse In)

Lock-in / Field Sweep (used during CWNMR operation of the spectrometer)

Not used, no connections.

You can now switch **on** the POWER (feel for switch at rear left of the '**Mainframe**') - the POWER LED (red) on the **PS2 CONTROLLER** should light up.

Adjust the RIGHT and LEFT FEEDBACK LOOP potentiometers until the ERROR SIGNAL INDICATORS (LEDs) become dark (red means the potentiometer is set too high, green means too low; the correct setting is between red and green). When the LEDs are dark switch each FEEDBACK LOOP to CLOSED

'Push to Select' knob operation (Synthesizer and Programmer)

To adjust F (frequency, **Synthesizer**) and τ (tau, **Programmer**):

- 1) Rotate the knob to select (flashing cursor, top line of display) the parameter to be adjusted and push the knob in until a beep indicates that the parameter has been selected (flashing cursor moves to second line of display)
- 2) Rotate the knob to position the cursor on the number (second line of display) to be changed, then push the knob in (beep) to select the number (the selected number is now underlined)
- 3) Rotate the knob to change number to the desired value, then push the knob in (beep) to lock the value in
- 4) Rotate the knob (cursor now moves) to select the next number to adjust (i.e. back to step 2), or to select (up arrow) which takes the cursor back to the top line of the display

To adjust A (A_len), B (B_len), N (Num_B) and P (Period = repetition time of pulse sequence) on the **Pulse Programmer**, only perform steps 1) and 3) above.

Using the 'Push to Select' knobs set the following:

F (**Synthesizer**): 21.70000 MHz
 A (A_len, **Programmer**): 2.50 μ s
 B (B_len, **Programmer**): 0.02 μ s
 N (Num_B, **Programmer**): 0
 P (Period, **Programmer**): 300.0 ms

'Tektronix' TDS 2002B DSO (Digital Storage Oscilloscope)

Switch on the **DSO** and recall Setup 1 as follows:

Press the SAVE/RECALL pushbutton (top right of DSO front panel)

Using the side-menu buttons (immediately to the right of the screen) select:

Action – Recall Setup, Recall From – Setup, Setup – 1, Recall (press)

The **DSO** screen should now look like Fig. 1. The signal displayed in CH 1 is called the **free induction decay or FID**.

To use cursors to perform on-screen measurements:

Press the CURSOR pushbutton (top right of DSO front panel)

Using the side-menu buttons (immediately to the right of the screen) select:

Type – Amplitude, Source – CH 1, Cursor 1, or Cursor 2

Use the 'Multipurpose' knob (rotating knob near top right of screen) to position the cursor selected. ΔV is the voltage difference between the cursors.

To print screen images via the PC:

Switch on the PC and open 'OpenChoice Desktop'

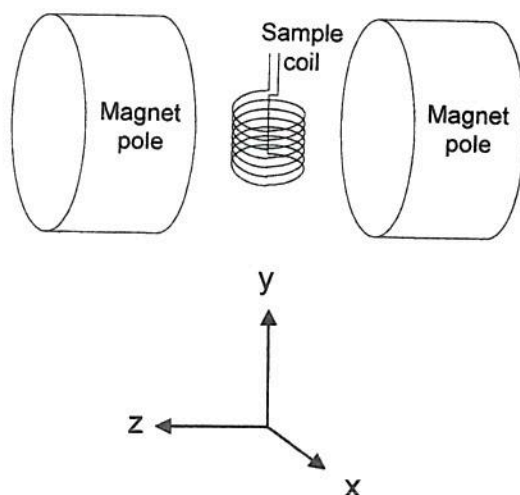
Press 'Select Instrument' and double click USB::

Press 'Get Screen' then 'Copy to Clipboard'

You can then paste the screen image into MS Word etc.

MAGNETIC FIELD GRADIENTS adjustment (L-OIL sample)

The magnetic field in this experiment is supplied by a permanent magnet. To make the magnetic field at the sample as uniform (homogeneous) as possible, four small coils, which can generate magnetic field gradients, are situated near the sample position. The field gradients are adjusted by varying the currents through the gradient coils, as follows:



Simplified diagram of the permanent magnet and the RF sample coil (showing x, y, z directions)

Adjust the X, Y, Z and Z^2 potentiometers (in that order) to lengthen the 'tail' of the FID in CH 1 on the DSO. Repeat the adjustment sequence 3 or 4 times to get the longest possible 'tail'. This adjustment gives the best **magnetic field homogeneity** attainable over the sample volume. With the L-OIL sample you should be able to get an FID signal length of at least about 20ms.

Synthesizer frequency (F) adjustment (L-OIL sample).

First adjust the 0.001 MHz digit to give the maximum amplitude for the CH 2 trace (see Fig. 2), then adjust the 0.00001 MHz digit to 'stop' the CH 2 trace on the screen (see Fig. 3) – the frequency of the oscillations of the CH 2 trace represents the frequency **difference** (= **beat frequency**) between the synthesizer frequency and the Larmor frequency of the protons in the sample. With further very careful adjustment, you can make the CH2 trace **similar in shape** to the FID in CH 1 (i.e. beat frequency = 0). At this point the **synthesizer frequency** is the same as the **Larmor frequency**!

The spectrometer frequency will need to be **checked and readjusted reasonably often** throughout the experiment. It will change relatively fast, shortly after switching the spectrometer on, then more slowly as the electronics and magnet warm up and stabilise.

THE UNIVERSITY OF NEW SOUTH WALES
SCHOOL OF PHYSICS
THIRD YEAR LABORATORY

PULSED NMR

**DSO screen images
to aid experiment set-up**

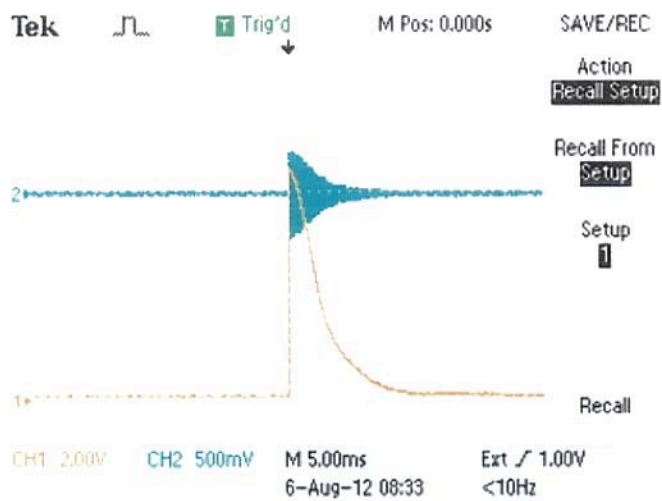


Fig. 1

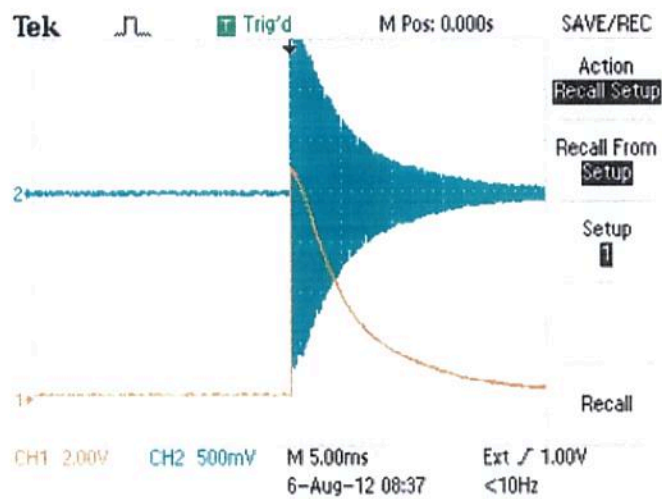


Fig.2

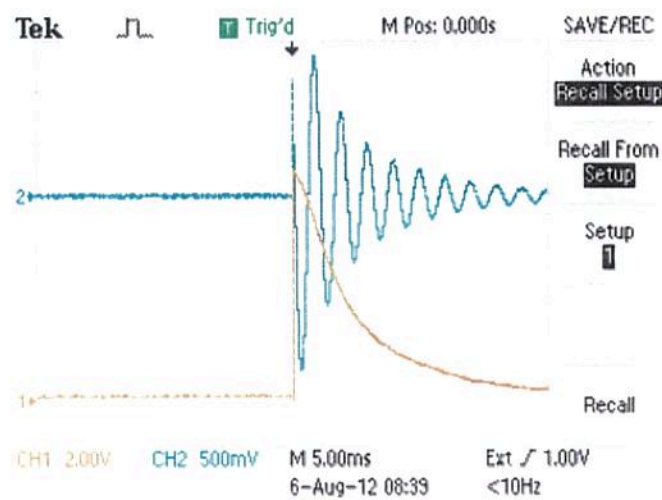
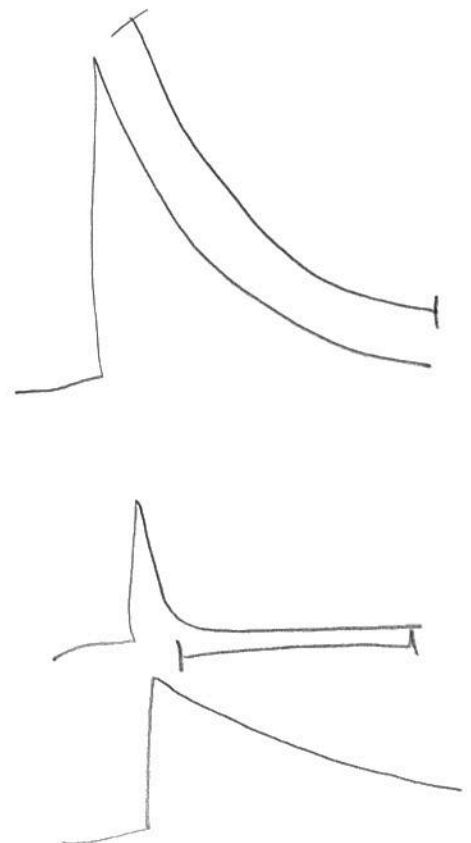


Fig. 3



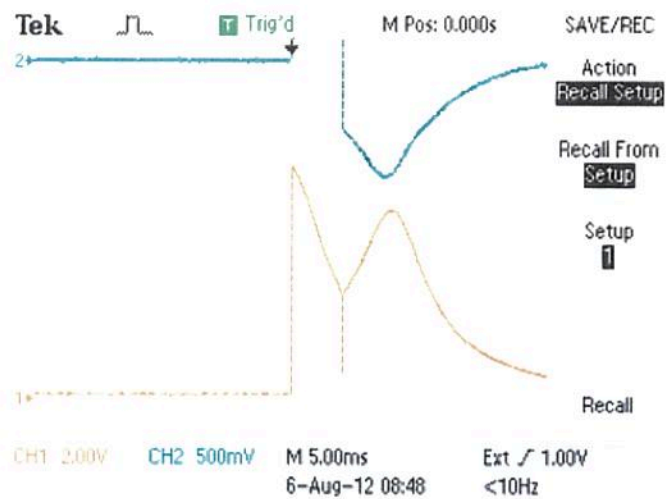


Fig. 4

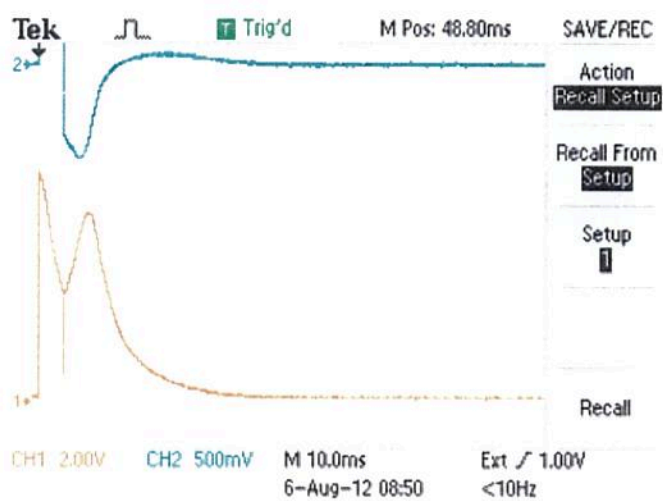


Fig. 5

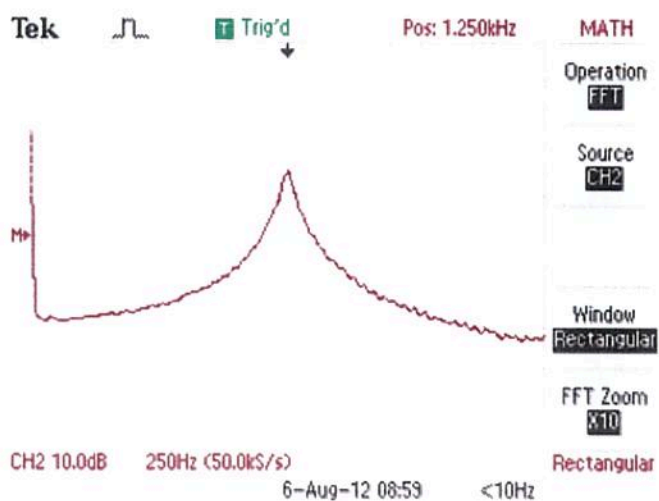


Fig. 6

Original

P N M R

Ref. 1

Theory

MEDICAL PHYSICS

MAGNETIC RESONANCE SPECTROSCOPY AND IMAGING

Prof. J. M. Pope, School of Physics, UNSW (1995)
TEXTBOOK

'The Physics of Medical Imaging', S. Webb (Ed.) Adam Hilger (1988).

REFERENCES

- 'Principles of Magnetic Resonance', C.P. Slichter, Springer-Verlag, (1978).
- 'Nuclear Magnetic Resonance', F.A. Rushworth and D. P. Tunstall, Gordon and Breach, (1973).
- 'Pulse and Fourier Transform NMR', T.C. Farrar and E.D. Becker, Academic Press, (1971).
- 'Experimental Pulse NMR, a Nuts and Bolts Approach', E. Fukushima and S.W. Roeder, Addison-Wesley, (1981).
- 'Fourier Transform NMR Techniques, a Practical Approach', K. Mullen and P.S. Pergosin, Academic Press, (1976).
- 'NMR Spectroscopy', R. K. Harris, Pitman, (1983).
- 'Paramagnetic Resonance', G.E. Pake, Benjamin, (1962).
- 'Practical NMR Imaging', M.A. Foster and J.M.S. Hutchison (Eds) IRL Press (1987).
- 'NMR and its Application to Living Systems', D.G. Gadian, O.U.P. (1982).
- 'Practical NMR Imaging', M.A. Foster and J.M.S. Hutchison (Eds) IRL Press (1987).
- 'NMR Imaging in Biomedicine', P. Mansfield and P.G. Morris, Academic Press (1982).
- 'NMR and its Application to Living Systems', D.G. Gadian, O.U.P. (1982).
- 'Nuclear Magnetic Resonance Imaging in Medicine and Biology', P.G. Morris, Oxford, Clarendon (1986).
- 'Biomedical Magnetic Resonance Technology', C.N. Chen and D.I. Hoult, Adam Hilger (1989).
- 'Principles of Nuclear Magnetic Resonance Microscopy', P. Callaghan, O.U.P. (1991).

REVIEW ARTICLES

- 'NMR Imaging' D. R. Bailes and D. J. Bryant, Contemp. Phys. 25, 441-475, (1984).
- 'A Unified Description of NMR Imaging, Data Collection Strategies and Reconstruction', K. F. King, Med. Phys. 11, 1-14, (1984).
- 'Magnetic Resonance Micro-imaging', J. M. Pope, Today's Life Science, 4, 40-46, (1992).
- 'Quantitative NMR Imaging of Flow', J. M. Pope and S. Yao, Concepts in Magnetic Resonance, 5, (October 1993).

MRI JOURNALS

- Magnetic Resonance Imaging (Pergamon).
- Magnetic Resonance in Medicine (Academic Press).
- Journal of Magnetic Resonance Imaging.
- Reviews of Magnetic Resonance in Medicine (Pergamon).
- Concepts in Magnetic Resonance.

1. INTRODUCTION

The phenomenon of nuclear magnetic resonance (NMR) was discovered independently by Bloch Hansen and Packard and by Purcell Torrey and Pound in 1946. Applications in chemistry followed with the discovery of the chemical shift in 1950, so that NMR has become a standard analytical tool in organic chemistry.

In the 1960's and 1970's NMR was increasingly applied to studies of biological tissues, tissue extracts, whole cell suspensions, perfused organs and ultimately whole animals. This was made possible by developments in technology, particularly the advent of high field and large bore superconducting magnets.

The first NMR images were demonstrated by Lauterbur in 1972. Magnetic Resonance Imaging (MRI) became a clinical imaging modality in the early 1980's and has developed rapidly since then to become a major diagnostic tool. There are now several thousand MRI installations worldwide.

In these lectures we will concentrate initially on the basic principles of NMR and then go on to outline the theory of MRI. However it is worth emphasising that, in addition to its ability to generate exquisitely detailed images of internal anatomy, NMR can provide detailed information at a molecular level on the physical and chemical environment and state of motion of a particular nuclear species. Some important applications of NMR therefore include:

1. Determination of molecular structure of organic molecules.
2. Measurement of molecular motion in liquids, liquid crystals, molecular crystals and solids.
3. Structure of proteins in solution.
4. Studies of biochemical processes 'in vivo'.
5. Magnetic Resonance Imaging (MRI).

In addition it is worth pointing out that the theory of electron spin resonance (ESR) has many parallels with that of NMR. The techniques differ in that, while NMR arises as a consequence of the magnetic properties of atomic nuclei, ESR involves the magnetism of atoms or molecules containing 'unpaired' electrons.

2. PRINCIPLES OF NUCLEAR MAGNETIC RESONANCE

Atomic nuclei carry a positive charge Ze . They also in most cases possess a further property which is essential if they are to exhibit magnetic resonance, that of spin angular momentum - i.e. the behave as if they are spinning about an axis. (In this respect they are similar to electrons).

The combination of positive charge and spin means that nuclei other than those with zero spin (such as ^4He , ^{12}C , ^{16}O , etc.) possess a magnetic moment:

$$\mu = \gamma \hbar I$$

where γ is the gyromagnetic ratio of the nucleus and $\hbar = h/2\pi$. Nuclear spin is quantised in units of $\hbar$. The magnitude of the spin angular momentum (in units of $\hbar$) is given by $|I| = [I(I+1)]^{1/2}$, where I is the spin quantum number, which can take either integral or half integral values $I = 0, 1/2, 1, 3/2, \dots$

A particular nucleus (isotope) can be characterised for NMR purposes by values for the gyromagnetic ratio γ and the spin angular momentum quantum number I .

Thus nuclei with non-zero spin are magnetic. If placed in a magnetic field $\mathbf{B}_0$ such a nucleus will experience a torque, $\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}_0$ tending to rotate the nucleus to align with the field direction (like a compass needle in the Earth's magnetic field). However there is an important difference from the case of a compass needle. The fact that the nucleus possesses angular momentum which must be conserved, means that its motion is to precess about the field direction (just like the axis of a top precesses about the Earth's gravitational field). The equation of motion of this **Larmor precession** is obtained by equating the torque to the resulting rate of change of angular momentum, viz.

$$\begin{aligned}\boldsymbol{\tau} &= \boldsymbol{\mu} \times \mathbf{B}_0 = \frac{d}{dt}(\hbar \mathbf{I}) \\ &= \frac{1}{\gamma} \frac{d\boldsymbol{\mu}}{dt} \\ \therefore \frac{d\boldsymbol{\mu}}{dt} &= \gamma \boldsymbol{\mu} \times \mathbf{B}_0\end{aligned}$$

This equation describes the precession of the nucleus about $\mathbf{B}_0$ with frequency

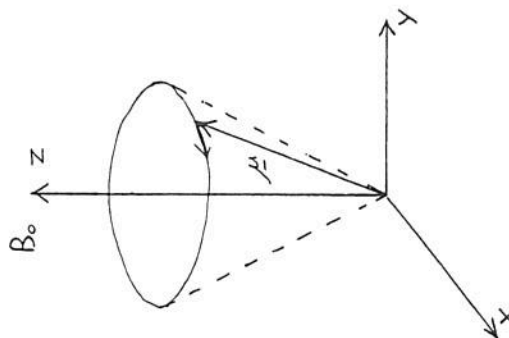
$$\omega_0 = -\gamma \mathbf{B}_0 \quad (1)$$

- the Larmor frequency.

Note - if we transform to a frame of reference rotating at ω about $\mathbf{B}_0$, the apparent rate of change of $\boldsymbol{\mu}$ in the rotating frame is given by (see Webb p394)

$$\begin{aligned}\frac{\delta \boldsymbol{\mu}}{\delta t} &= \frac{d\boldsymbol{\mu}}{dt} - \boldsymbol{\omega} \times \boldsymbol{\mu} \\ &= \gamma \boldsymbol{\mu} \times [\mathbf{B}_0 + \boldsymbol{\omega}/\gamma] \\ &= 0 \quad \text{for } \boldsymbol{\omega} = \omega_0 = -\gamma \mathbf{B}_0\end{aligned}$$

Clearly then if $\boldsymbol{\omega} = \omega_0 = -\gamma \mathbf{B}_0$ the magnetic moment $\boldsymbol{\mu}$ appears stationary in the rotating frame. Thus the motion of $\boldsymbol{\mu}$ in the laboratory frame is to precess about $\mathbf{B}_0$ at the Larmor frequency.



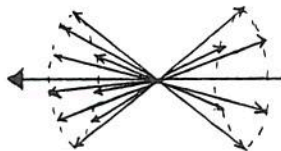
3. SAMPLE MAGNETISATION

In practice we cannot detect the magnetisation of a single nucleus. However in the presence of an applied field $\mathbf{B}_0$ the sample as a whole will acquire a net magnetisation $\mathbf{M}_0$ which in equilibrium will be aligned parallel to $\mathbf{B}_0$

$$\mathbf{M}_0 = (\chi_0/\mu_0)\mathbf{B}_0 = \chi_0 \mathbf{H}_0 \quad (2)$$

where χ_0 is the static susceptibility of the sample.

The static magnetisation $\mathbf{M}_0$ is the vector sum of the magnetic moments of the individual nuclei. In practice there will be a slight excess of nuclei with components of magnetic moment parallel to $\mathbf{B}_0$ rather than antiparallel, giving rise to the net magnetisation $\mathbf{M}_0$ parallel to $\mathbf{B}_0$. In equilibrium there will be no component of magnetisation perpendicular to $\mathbf{B}_0$.



If we perturb the magnetisation such that $\mathbf{M}$ is no longer parallel to $\mathbf{B}_0$ the magnetisation as a whole will precess about $\mathbf{B}_0$ according to the equation

$$d\mathbf{M}/dt = \gamma \mathbf{M} \times \mathbf{B}_0 \quad (3)$$

Such a precessing magnetisation, of angular frequency ω_0 can be detected by means of a coil wrapped around the sample (or patient).

4. ROTATING FRAME AND EFFECTIVE FIELD

It is instructive to view the precessing (non-equilibrium) magnetisation $\mathbf{M}$ from a frame of reference rotating with angular frequency ω_0 about the direction of $\mathbf{B}_0$, which we shall henceforth designate the z direction. In this rotating reference frame (x', y', z) the nuclear moments and hence the net magnetisation appears stationary. It is as if the effect of the magnetic field had been transformed away.

In general it can be shown (see Webb p.395) that in a frame of reference rotating at angular frequency ω about $\mathbf{B}_0$ the magnetisation obeys the general equation:

$$\delta \mathbf{M}/\delta t = \gamma \mathbf{M} \times \mathbf{B}_e \quad (4)$$

where $\delta \mathbf{M} / \delta t$ represents the apparent rate of change of magnetisation in the rotating frame and the 'effective field' in this frame:

$$\mathbf{B}_e = \mathbf{B}_0 + \omega / \gamma \rightarrow 0 \quad \text{for } \omega = \omega_0 = -\gamma \mathbf{B}_0$$

5. EFFECT OF RF PULSES

Consider now the effect of applying, in addition to the static magnetic field $\mathbf{B}_0$ along $\hat{z}$, a circularly polarised field of frequency ω with magnetic induction vector $\mathbf{B}_1$ rotating in the x, y plane in the same sense as the precessing magnetisation.

In the rotating frame, this rotating field will appear static. If we choose the x' axis of the rotating frame along $\mathbf{B}_1$, then the effective field in the rotating frame becomes

$$\begin{aligned} \mathbf{B}_e &= (\mathbf{B}_0 + \omega / \gamma) + \mathbf{B}_1 \\ &= (B_0 + \omega / \gamma) \hat{z} + B_1 \hat{x}' \end{aligned} \quad (5)$$

and if $\omega = \omega_0 = -\gamma B_0$ this reduces to

$$\mathbf{B}_e = B_1 \hat{x}'$$

i.e. the magnetisation in the rotating frame 'sees' only the static vector $\mathbf{B}_1$.

The motion of the magnetisation vector in the rotating frame then will be to precess about $\mathbf{B}_1$ with angular frequency $\omega_1 = -\gamma B_1$ according to the equation

$$\delta \mathbf{M} / \delta t = \gamma \mathbf{M} \times \mathbf{B}_1$$

Hence if the magnetisation is initially in equilibrium along the $\hat{z}$ direction before the rotating field is turned on, ($\mathbf{M} = M_0 \hat{z}$), it will precess round in the $y'z$ plane. If the rotating field is applied in the form of a pulse of duration τ_p , then the angle of precession

$$\theta_p = \omega_1 \tau_p = \gamma B_1 \tau_p \quad (6)$$

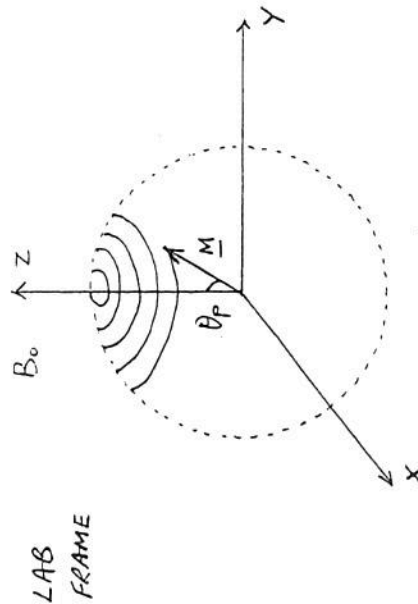
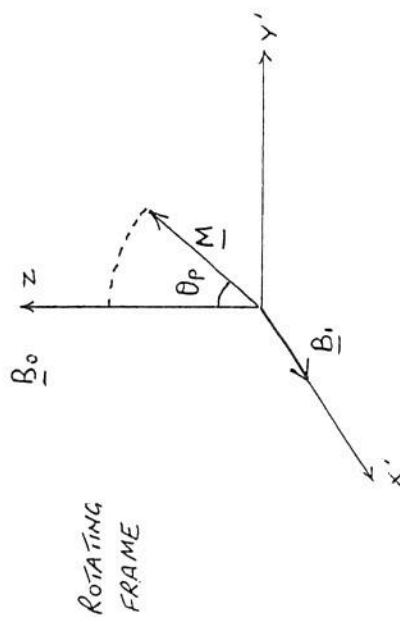
Thus by appropriate choice of τ_p (or B_1) we can select $\theta_p = \pi/2$ (a 90° pulse) or $\theta_p = \pi$ (a 180° pulse). The latter pulse has the effect of completely inverting the magnetisation.

In the laboratory frame the motion appears more complex, the tip of the magnetisation vector following a spiral path on the surface of a sphere. However the overall effect is the same, the net sample magnetisation being tipped through an angle θ_p with respect to the $\mathbf{B}_0$ direction.

NOTE Commonly in pulsed NMR an oscillating, (plane polarised), rather than rotating, (circularly polarised), RF field of the form $\mathbf{B}_1(t) = 2B_1 \cos \omega t \hat{x}$ is applied. This can be decomposed into two counter-rotating components each of amplitude half that of the oscillating field:

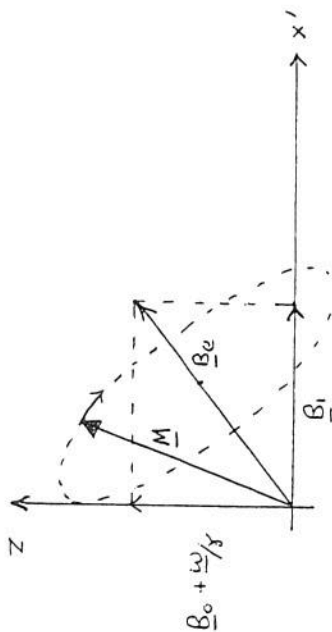
$$\mathbf{B}_1(t) = B_1(\hat{x} \cos \omega t + y \sin \omega t) + B_1(\hat{x} \cos \omega t - y \sin \omega t).$$

One component, which rotates in the opposite sense to the precessing magnetisation and is therefore $2\omega_0$ off-resonance, can effectively be ignored.



6. RESONANCE CONDITION

More generally if $\omega \neq \omega_0$, the magnetisation will precess about $\mathbf{B}_e$ following the surface of a cone, according to equation 4:



It is important to note that the magnetisation will only deviate significantly from the z direction if $\mathbf{B}_e$ itself subtends a large angle with respect to $\mathbf{B}_0$. This will only occur if $B_0 + \omega/\gamma$ is small, i.e. $\omega \ll \omega_0 = -\gamma B_0$. NMR is a resonance phenomenon, the magnitude of the resonant frequency being given by the resonance condition:

$$\nu_0 = \omega_0/2\pi = \gamma B_0$$

c.f. equation (1), where here $\gamma = \gamma/2\pi$ and ν_0 is in Hz.

Ex for a field of 1 Tesla:

Nucleus	Resonant Frequency (MHz)
^1H	42.6
^3P	17.2
^{13}C	10.7
^{23}Na	11.3
^{19}F	(40.1)

Typical NMR frequencies are therefore of order of tens to hundreds of MHz - ie in the radiofrequency (RF) range.

In MRI the ^1H nucleus has been used almost exclusively to date. The signals derive mainly from water, H_2O and fat, $(\text{CH}_2)_n$. However ^3P can yield very useful information about tissue energy metabolism and pH, giving rise to the related technique of in-vivo magnetic resonance spectroscopy (MRS).

7. BLOCH EQUATIONS

It is clear that equation (4) is incomplete in that it predicts that in the absence of an RF field, for which $\mathbf{B}_e \rightarrow 0$ in a frame of reference rotating at the resonant frequency ω_0 , the magnetisation appears static - ie in the laboratory frame it will precess forever about $\mathbf{B}_0$. This clearly ignores the effects of relaxation which ensure that following any perturbation the magnetisation must return to equilibrium along $\mathbf{B}_0$. Bloch adopted a phenomenological approach by adding to equation (4) terms to take account of this relaxation. These terms must take account of both:

- Recovery of the z component of magnetisation M_z back to its equilibrium value M_0 (in the absence of a perturbing RF field) with time constant T_1 , the longitudinal or spin-lattice relaxation time; and
- Decay of the transverse components of magnetisation M_x and M_y over a time scale characterised by the transverse or spin-spin relaxation time T_2 .

Bloch made the assumption that these relaxation processes would be characterised by exponential relaxation functions. While this is generally true for longitudinal relaxation and for transverse relaxation in liquids, it is not generally the case for spin-spin relaxation in solids.

Thus Bloch added to equation (4) the following additional terms to take account of the effects of these relaxation processes on the corresponding components of magnetisation:

$$\begin{aligned}\delta M_x/\delta t &= -M_x/T_2 \\ \delta M_y/\delta t &= -M_y/T_2 \\ \delta M_z/\delta t &= (M_0 - M_z)/T_1\end{aligned}$$

Overall the Bloch equations in the rotating frame then become:

$$\begin{aligned}\delta M_x/\delta t &= M_y(\omega - \omega_0) - M_x/T_2 \\ \delta M_y/\delta t &= -M_x(\omega - \omega_0) - M_z\omega_1 - M_y/T_2 \\ \delta M_z/\delta t &= M_y\omega_1 + (M_0 - M_z)/T_1\end{aligned}\quad (7)$$

where we have made use of equations (1), (5) and

$$\omega_1 = -\gamma B_1 \quad (8)$$

In the laboratory frame ($\omega = 0$) and in the absence of an RF pulse ($\omega_1 = 0$) so that $B_e \equiv B_0$ these reduce to:

$$\begin{aligned}\delta M_x/\delta t &= -M_y\omega_0 - M_x/T_2 \\ \delta M_y/\delta t &= M_x\omega_0 - M_y/T_2 \\ \delta M_z/\delta t &= (M_0 - M_z)/T_1\end{aligned}$$

which have solutions

$$\begin{aligned}M_x(t) &= [M_x(0) \cos(\omega_0 t) + M_y(0) \sin(\omega_0 t)]e^{-t/T_2} \\ M_y(t) &= [M_x(0) \sin(\omega_0 t) - M_y(0) \cos(\omega_0 t)]e^{-t/T_2} \\ M_z(t) &= [M_z(0) - M_0]e^{-t/T_1} + M_0\end{aligned}$$

Thus for example following a 90° pulse which tips the magnetisation M_0 along the y' axis in the rotating frame

$$\begin{aligned} M_x(t) &= M_0 \sin(\omega_0 t) e^{-t/T_2} \\ M_y(t) &= -M_0 \cos(\omega_0 t) e^{-t/T_2} \\ M_z(t) &= M_0 (1 - e^{-t/T_1}) \end{aligned} \quad (9)$$

Note that in general $T_1 > T_2$. While recovery of the z component of magnetisation must be at the expense of transverse magnetisation, additional processes contribute to T_2 which are not involved in T_1 . (We will see that in many cases the transverse or spin-spin relaxation time T_2 is governed by interactions between the spins).

8. THE FREE INDUCTION DECAY (FID) AND NMR SPECTRUM

The above results imply that the transverse magnetism $M_T = \sqrt{M_x^2 + M_y^2} = M_0 e^{-t/T_2}$, (which initially ($t = 0$) is equal to M_0).

- (i) rotates (precesses) about B_0 with frequency ω_0
- (ii) decays exponentially with time constant T_2 .

This precessing magnetisation will induce a voltage in the RF coil used both to transmit the RF pulses and detect the resulting signal. The coil forms part of a high Q tuned circuit, so that the induced signal following a 90° pulse is given by [King and Moran Med. Phys. 11, 1-14 (1984)]:

$$S(t) = A Q \omega_0 \eta V_c M_0 e^{-t/T_2} \cos \omega_0 t$$

where A is an amplification factor, V_c is the RF coil volume and η a 'filling factor'.

This decaying oscillatory signal at frequency ω_0 is called the free induction decay (FID).

More generally for a signal of arbitrary phase we write:

$$S(t) = A Q \omega_0 \eta V_c M_0 e^{-t/T_2} e^{i\omega_0 t} \quad (10)$$

since $e^{i\omega_0 t} = \cos \omega_0 t + i \sin \omega_0 t = e^{i2\pi\nu_0 t}$.

The frequency spectrum of this decaying transient signal is given by Fourier transformation. We define Fourier transform pairs.

$$F(\nu) = \int_{-\infty}^{\infty} f(t) e^{-i2\pi\nu t} dt \quad (11)$$

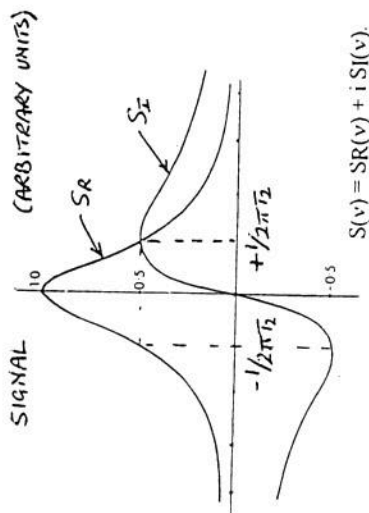
$$f(t) = \int_{-\infty}^{\infty} F(\nu) e^{+i2\pi\nu t} d\nu$$

Then since $S(t)$ is non-zero only for $t > 0$

$$\begin{aligned} S(\nu) &= A Q 2\pi\nu_0 \eta V_c M_0 \int_0^{\infty} e^{-t/T_2} e^{i2\pi(\nu_0 - \nu)t} dt \\ &= A Q 2\pi\nu_0 \eta V_c M_0 T_2 \{ (1 + i2\pi(\nu_0 - \nu)T_2) / (1 + [2\pi(\nu_0 - \nu)T_2]^2) \} \end{aligned} \quad (12)$$

i.e a Lorentzian

Thus in a uniform field, the NMR signal is sharply peaked at frequencies close to the Larmor frequency ω_0 . Note the real part of $S(\nu)$ (the NMR spectrum or absorption signal) has a FWHM equal $1/\pi T_2$ in Hz.



The NMR spectrum is a plot of the relative numbers of nuclei precessing at a given frequency, versus frequency. (It can be measured directly in a continuous wave NMR experiment by applying a low level RF field B_1 to the nuclear spin system, sweeping its frequency slowly through resonance and recording changes in the transverse magnetisation or absorption of energy by the spin system). In the 'ideal' case of an assembly of non-interacting spins of a particular isotope subject to a perfectly uniform static field B_0 , all the nuclei would precess at the same rate and the spectrum would reduce to a single δ function. However in practice interactions between the spins and with their molecular environment give rise to different spins in the sample experiencing slightly different effective local fields. As a result different spins in the sample precess at slightly different frequencies, resulting in broadening and/or splitting of the NMR spectrum into discrete resonance peaks. This structure in the spectrum in turn is largely responsible for its information content, since the interactions yield information about molecular structure and bonding.

Eventually the magnetisation must return to equilibrium in the direction of B_0 . Consequently there will be contribution to the decay of transverse magnetisation and hence T_2 from spin lattice relaxation. For this reason T_1 represents an upper limit to the decay of transverse magnetisation. For non-viscous liquids $T_2 \approx T_1$ while for rigid solids in which spin-spin interactions are strong, $T_2 \ll T_1$.

In the foregoing we assumed that the applied static magnetic field B_0 was homogeneous across the sample. In practice it will never be perfectly so, but will exhibit some non-uniformity $B_z = B_0 \pm \delta$. Consequently spins in different parts of the sample will precess at slightly different frequencies.

Viewed from a frame of reference rotating at the average Larmor frequency ω_0 , following a 90° pulse the magnetisation will de-phase. Thus in practice the apparent decay of transverse magnetisation will occur with time constant $T_2^* < T_2$ i.e. more rapidly than the intrinsic transverse relaxation processes dictate. Alternatively and equivalently the NMR spectrum will be broadened, exhibiting a FWHM given by $1/\pi T_2^*$ rather than $1/\pi T_2$.

10. THE CHEMICAL SHIFT

One factor which gives rise to differences in precession frequency between different spins in a sample is the 'chemical shift'. This arises because nuclei in a given (diamagnetic) sample are partially screened from the full effect of the applied static magnetic field by the electron cloud of the atom or molecule of which they form part. For liquid samples this screening is isotropic, (averaged by the rapid molecular tumbling), so that the field experienced by a particular nucleus:

$$B = B_0 (1 - \sigma) \quad (13)$$

where σ , which is dimensionless, is the chemical shielding constant. The corresponding precession frequency for this nucleus:

$$\omega = \gamma (1 - \sigma) B_0 \quad (14)$$

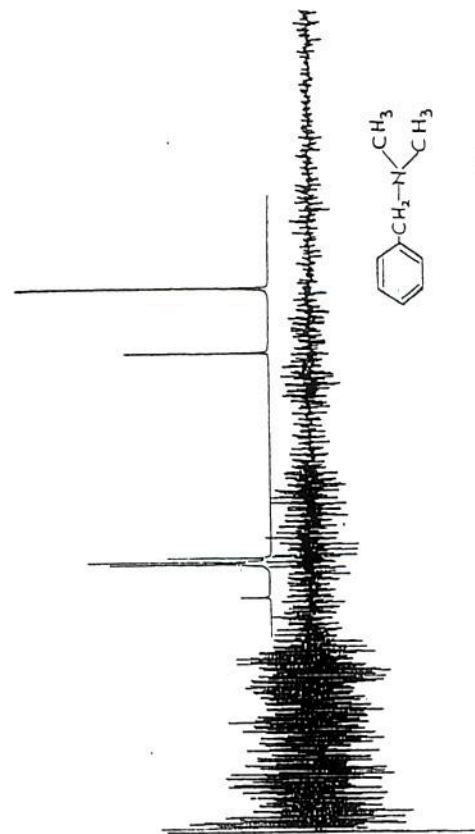
In practice the extent of this chemical shielding is very small, typically only corresponding to a few parts per million (ppm) of the overall static field for ^1H and ranging up to a few hundred ppm for nuclei such as ^{13}C or ^3P . However it varies slightly for nuclei in different electronic environments - i.e. different chemical groups so that the form of the NMR spectrum reflects the structure of the molecules in the sample. As a result, NMR has become a standard analytical tool for molecular structure determination in many branches of chemistry and biochemistry.

Clearly for two groups of nuclei of the same atomic species, but different chemical shielding constants σ_1 and σ_2 , the difference between their (average) precession frequencies in a uniform static field B_0 :

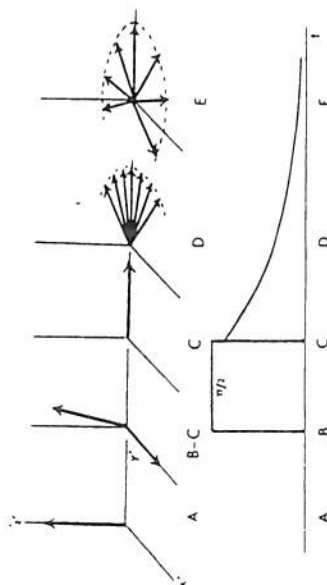
$$\Delta\omega = \gamma (\sigma_2 - \sigma_1) B_0$$

- i.e. the spectral resolution scales with the static field strength. Conventionally, however, chemical shifts are expressed with respect to some standard reference substance, (e.g. tetramethylsilane (TMS) for ^1H and ^{13}C). As a fraction of the reference field/frequency, the chemical shift δ is then field independent and given in ppm by:

$$\delta = (\Delta\omega/\omega_{\text{ref}}) \times 10^6 \approx (\sigma_{\text{ref}} - \sigma) \times 10^6 \quad (15)$$



The ^{13}C FID and frequency spectrum of N,N' -dimethylbenzylamine (^1H decoupled)



9. PHASE SENSITIVE DETECTION

In general in NMR, signals are detected with two phase coherent demodulators which are adjusted to have their reference phases at 90° to each other (quadrature detectors). If the reference phase is adjusted correctly these correspond to the real and imaginary components of $S(t)$. A complex Fourier transform then yields the real (ν mode) and imaginary (ω mode) components of $S(\nu)$. It can be shown (see e.g. Harris p. 70) that these correspond respectively to the out of phase (M_x) and in-phase (M_y) components of transverse magnetisation in the rotating frame.

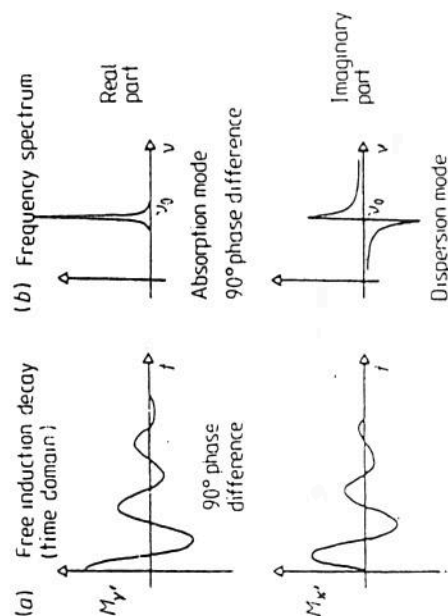


Figure 8.11 (a) The FID signals obtained following quadrature detection, and (b) the absorption and dispersion mode signals obtained after Fourier transformation.

11. MEASUREMENT OF RELAXATION TIMES

We have seen that the longitudinal or spin-lattice relaxation time T_1 characterises the recovery of components of magnetisation parallel to the static magnetic field B_0 , following a perturbation. However the magnetic resonance experiment is sensitive only to magnetisation components perpendicular to B_0 . To measure T_1 then we use a pulse sequence which firstly perturbs the z magnetisation by applying either a 90° or 180° pulse, after which the magnetisation is allowed to recover for a time τ . Following this delay the magnetisation remaining in the B_0 direction is then 'read out' by applying a second pulse of 90° tip angle and recording the FID. In the case of the 90° - τ - 90° (saturation recovery) sequence the first pulse equalises the populations of the states (corresponding to infinite spin temperature - see below). During the relaxation delay τ the transverse magnetisation dephases and the longitudinal component recovers, such that the FID amplitude (or that of any spectral component) following the second 90° pulse is given by:

$$M_z(\tau) = M_0(1 - e^{-\tau/T_1}) \quad (16a)$$

and the form of the FID, at least in liquid-like samples and biological materials can typically be approximated by:

$$M_y(t) = M_z e^{-t/T_2^*} \quad (16b)$$

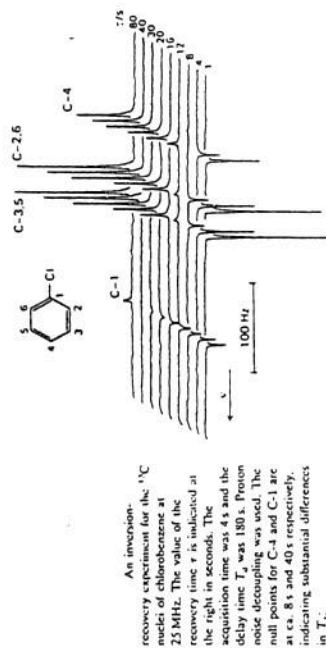


Exponential increase in size of free induction decay in 90° - τ - 90° sequence with increasing τ .

The 180° pulse of the 180° - τ - 90° (inversion-recovery) sequence reverses the initial state populations and sample magnetisation, corresponding to a state of negative spin temperature. No transverse magnetisation is generated by the 180° pulse, so there is no free induction signal following it. The z magnetisation, which determines the amplitude of the FID signal following the 90° pulse, then recovers according to

$$M_z(\tau) = M_0(1 - 2e^{-\tau/T_1}) \quad (17)$$

Note that in this case M_z passes through zero when $\tau = T_1 \ln 2$.

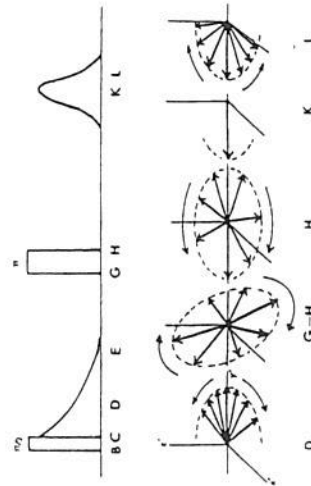


For solid samples and viscous liquids, with short intrinsic transverse relaxation times T_2 , the spin-spin relaxation function can in principle be determined directly from the FID or spectrum, although there may be experimental problems associated with receiver 'dead time' following the 90° excitation pulse. In the case of non viscous liquids in which the FID signal is affected by magnetic field inhomogeneities leading to a transverse relaxation characterised by a time constant T_2^* (determined by the B_0 field profile and the sample shape), the true intrinsic spin-spin relaxation time T_2 can be measured by a 'spin echo' method. This technique relies on the fact that following a 90° pulse, the contribution to de-phasing of the transverse magnetisation from inhomogeneities in the B_0 field is reversible. Application of a 180° pulse at time τ following the initial 90° pulse causes a reversal of the de-phasing produced by variation in the static field experienced by different parts of the sample, leading to the formation of a 'spin echo' at time 2τ , (see diagram). Intrinsic spin-spin relaxation processes are not reversible so that the magnitude of the echo signal will decay as the pulse spacing τ is increased.

$$M(2\tau) = M_0 e^{-2\tau/T_2} \quad (18)$$

leading to a method of measuring T_2 in an inhomogeneous magnetic field. However it should be noted that motion of the spins resulting from molecular translational diffusion will also give rise to incomplete re-phasing of the transverse magnetisation and decay of the spin echo, so that for a liquid of low viscosity (high diffusion coefficient D) in a field gradient G there will be two contributions to the echo decay.

$$M(2\tau) = M_0 e^{-2\tau/T_2} e^{-2\gamma^2 G^2 D \tau^3} \quad (19)$$



However, when searching for a signal in the pulsed mode, it is essential that the coil be on, or nearly on, resonance with the RF frequency, and that the RF frequency be at, or nearly at, the precession frequency of the nuclei being measured. This is not difficult to achieve, since the magnet has the Larmor precession frequency of a proton sample written on the serial label. This label appears on both the magnet and the electronics Mainframe. Of course, this is not an exact number since the magnetic field in the gap is temperature dependent. It is, however, close enough for a student to see a signal if the RF sample probe is properly tuned to that frequency. It is very difficult to observe a signal if the sample coil is not tuned near the RF frequency of the synthesizer.

I.2 Pickup Probe

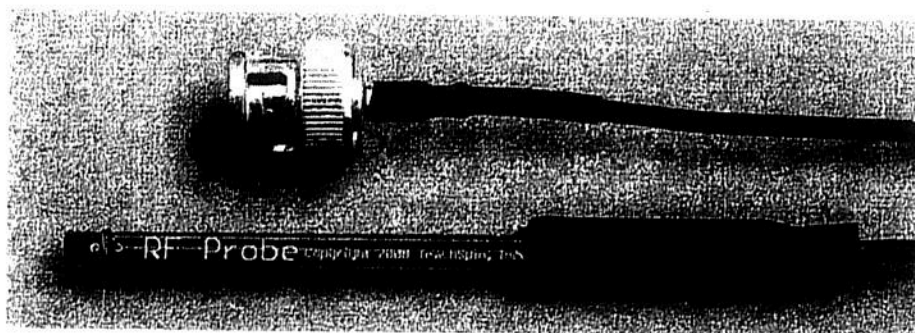


Figure 2.11 RF Pickup Probe

Figure 2.11 shows a diagram of the RF pickup probe. As described above, this probe is essential in the early stages of tuning the RF sample probe to the Larmor precession frequency. There are no test points or pickoff points in the RF circuits, so we provided the user with this probe to examine the RF field DURING THE PULSE. The oscillating RF field in the coil produces an emf in the coil, and that emf, in turn, generates an oscillating voltage across the 50 Ω load. This voltage can easily be observed on an oscilloscope, as shown in Figure 2.12.

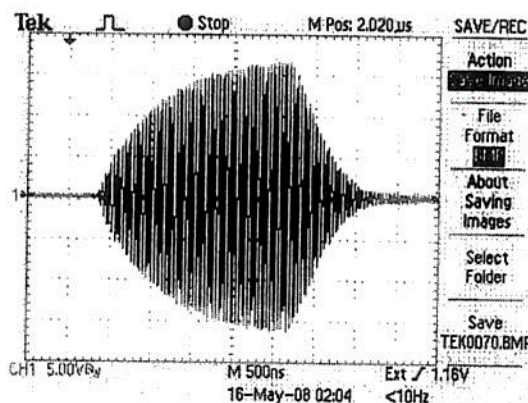


Figure 2.12 Pickup Signal During RF pulse

This voltage can be optimized primarily by adjusting the two tuning capacitors (25 – 150 pf compression, 1 – 30 pf piston) using the spectral tool provided. **BE CAREFUL NOT TO USE A STEEL SCREW DRIVER – THIS AREA HAS A LARGE MAGNETIC FIELD. YOU CAN DAMAGE THE MAGNET**

I. RF SAMPLE PROBE HEAD

I.1 Overview

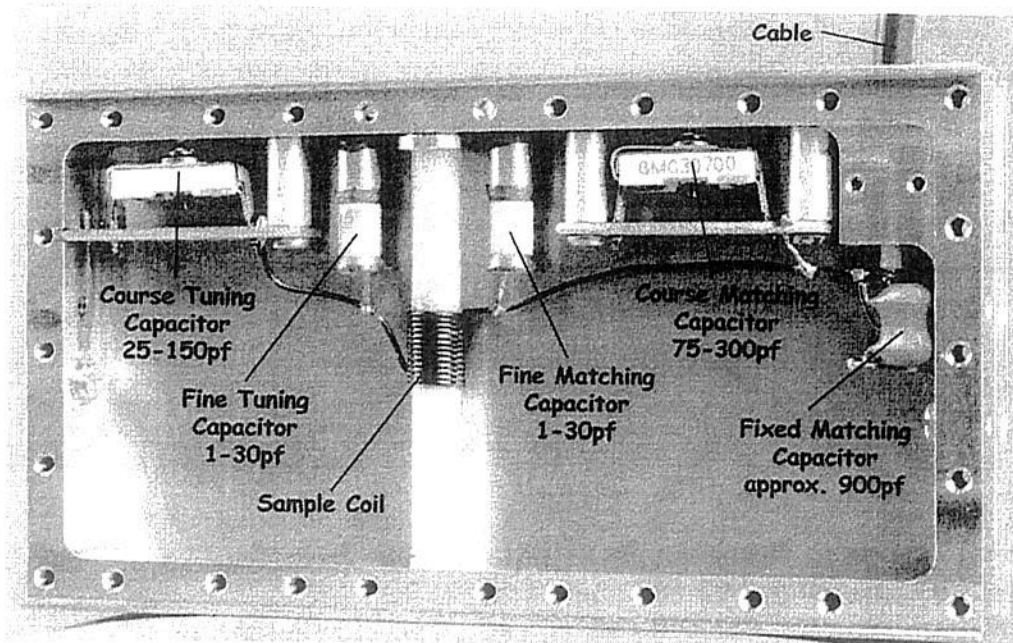


Figure 2.9 RF Sample Probe (RF Shield and Modulation and Gradient Coils not Shown)

The single coil RF sample probe is shown in Figure 2.9, without the RF shield and the modulation and gradient coils which form the side walls. The circuit diagram is shown in Figure 2.10.

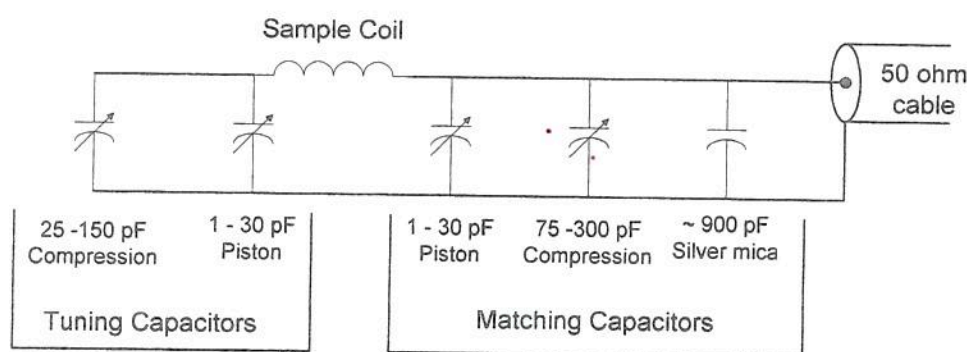


Figure 2.10 Circuit Diagram for RF Sample Probe

The two capacitors on the left side combine to tune the sample coil into series resonance at the frequency of the applied RF signal. This tuning can be observed with the “pick-up probe” that is inserted into the sample chamber. The RF field in the coil produces an emf in the two loops of the probe and that emf can be observed on an oscilloscope. Using this pick-up signal, the user can tune the sample coil to resonance by adjusting both the compression and the piston capacitors on the tuning side. Changing the impedance matching capacitors (piston, compression, and fixed 900 pf) has a small effect on optimizing the RF power to the coil, but is important when using the spectrometer in CW mode. Since there is always at least 976 pf of matching capacitance, the sample probe is usually well enough matched for most pulsed NMR experiments.

Sample

no adjust

no adjust

Coarse

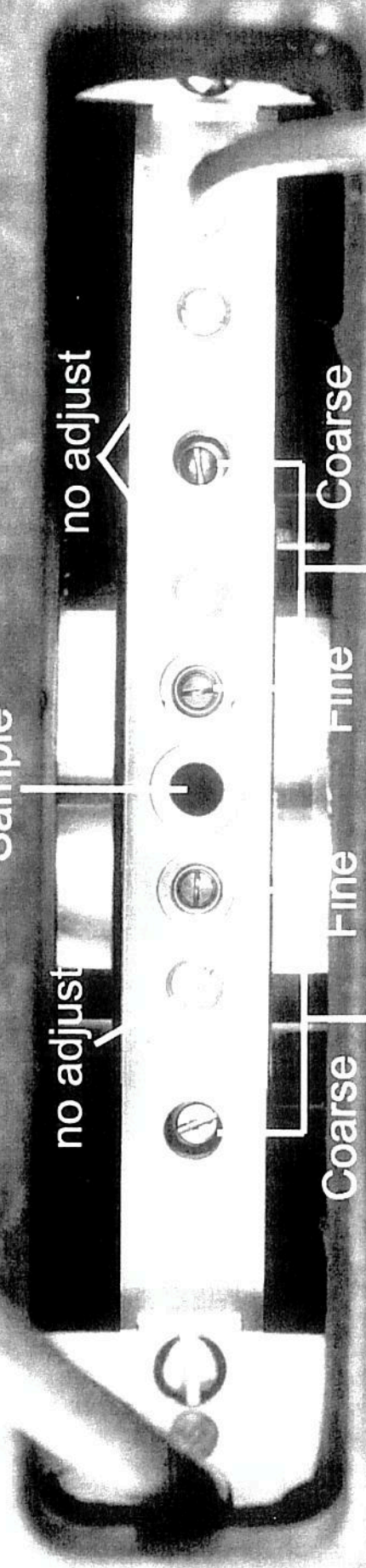
Fine

Fine

Coarse

Tuning
Capacitors

Matching
Capacitors



Optimization of the field gradients is achieved by systematically adjusting one gradient field after another while observing the Free Induction Decay (FID) envelope after a single 90° pulse - the longer the envelope, the more homogeneous the field. As good as they are, the gradient coils are not perfect. When the current is changed in the y-gradient coil, it may have some small effect on both the x and z linear gradients and may even produce a small effect on the value of the total field. Thus, one must go through the entire process several times to achieve the optimization of the field homogeneity. It is essential to optimize the field on a proton signal where the chemical shifts are small, and then use that homogenized field to observe the inequivalent fluorines in various fluorine samples.

Students should be able to obtain 5 ms T_2^* for the proton signal in light mineral oil. It may take some "fiddling around" to achieve this. Gradients can be added in either direction simply by flipping the reversing switch on the front panel. The currents in each coil can be monitored at the BNC connection under each control potentiometer. This point monitors the voltage across a 2.5 Ω precision resistor in series with the X and Y gradient coil, and a 1.25 Ω resistor for the Z and Z^2 gradient coils. These monitor points can be used to record the currents and to determine their stability as well as to trouble shoot any problems that might occur in the electronics.

Specifications:

$$\frac{\partial B_z}{\partial z} = 6.6 (\mu T / mm) / amp$$

$$\frac{\partial^2 B_z}{\partial z^2} = 20 (\mu T / mm^2) / amp$$

$$\frac{\partial B_z}{\partial x}, \frac{\partial B_z}{\partial y} = 7.1 (\mu T / mm) / amp$$

G. MAGNET YOKE ADJUSTMENT

The adjustment about to be explained should only be made by the faculty or staff in charge of this apparatus. *Students should not attempt this adjustment!*

The magnet pole pieces are adjusted at the factory to reduce or nearly eliminate the X, Y linear gradients over the sample. After you have optimized the magnetic field homogeneity over the sample by adjusting the electric current gradient coils, you should notice that the X and Y linear gradient current settings are at only a small fraction of their entire range. But suppose, after many years, or after the magnet has been moved several times, students discover that they could improve the homogeneity of the field even more, but they have "run out of current" in say the Y-linear gradient. If indeed this is correct, realigning the pole pieces will greatly reduce the inherent Y-gradient of the magnet. (Please Note - This realignment of the pole pieces will primarily affect X and Y gradients and has only a very small effect on the Z-gradients).

The procedure is as follows: Place a small (few drops) light mineral oil sample in the RF sample probe. Be sure it is in the proper location. Tune the spectrometer for an FID following a single 90° pulse. Turn down the gradient coil currents to zero. Insert the 3/32" hex wrench with T-handle, into one of the three 1/8" holes in the front heat sink and engage one of the three set screws. These set screws (which were initially adjusted at the factory) press on the front pole face and are used to align the front pole face to be parallel to the rear pole face. It is the "parallelness" of the two pole pieces that determines the X and Y linear gradients.

little as 10 Hz can be detected. This drift can be monitored using a two channel scope to compare the envelope output with the output of either of the phase sensitive detectors. (The phase and gain must be adjusted so that the two signals are equal.)

The controller can be operated for an indefinite time as long as the ambient temperature of the room is stable to about ± 1 C. Once the experiments are complete, the thermal loops on both of the magnetic loops should be switched to OPEN to allow the magnetic to establish equilibrium with the current room temperature. The field gradient coils should be left ON.

F. MAGNETIC FIELD GRADIENT COILS

The magnetic field homogeneity over the sample volume can be significantly enhanced by using a set of four coils that form the sides of the RF probe unit. These coils are designed to produce magnetic field **gradients** in the region around the sample when steady dc current is passed through them. There are four sets of coils producing linear field gradients in the x, y, and z directions and a quadratic field gradient in the z direction. (See Figure 2.8 for the definition of the x,y,z axes.)

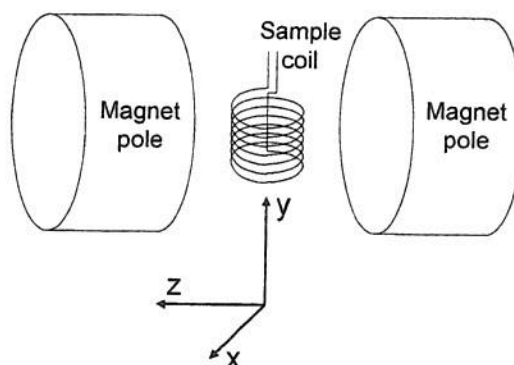


Figure 2.8 Simplified drawing of Magnetic Poles and Sample Coil showing x, y, and z axes.

Each gradient coil has two windings, one on each side of the RF probe. The coils are made on electronic circuit boards by electrochemical etching, and are therefore extremely dimensionally stable. Each is connected to a constant current regulated supply whose current is adjusted by a 10 turn potentiometer. The potentiometer shafts can be locked on the dial counter after adjustment. Current regulated supplies are used so that the current, which creates the field gradient, will not change as the resistance of the coils and leads changes with temperature.

For optimum field homogeneity, the field gradient coils should be adjusted by the students for each experiment. The gradients in the magnet may drift slightly with time and temperature, so the adjustment should be made before any critical experiment is performed. Please note that many experiments do not require optimum field homogeneity.

One can adjust the gradients with almost any liquid sample containing protons that has a "reasonable" T_1 and a $T_2 > 30$ msec. By reasonable we mean 30 – 100 msec, so that the repetition time can be 0.2 to 1.0 sec. Although ordinary water has a $T_2 \gg 30$ msec, the long repetition time required (since $T_1 \sim 2$ s) makes the experiment time consuming and cumbersome. Most light mineral oil samples will serve this purpose. T_2 can easily be estimated using a two-pulse sequence and observing the spin-echo.

without damage to either the coils or controller. Keeping the fields of the shim coils on requires that the spectrometer also be left on. However, we recommend that both the A and B pulses on the pulse programmer be turned off.

Gradient coils ON and loop OPEN is the best way to leave the apparatus during a semester, where it is important to obtain optimum field stability in the shortest time. In this configuration, the magnet comes to steady state equilibrium in the shortest time with its ambient temperature.

The objective is now to lock the temperature of each magnetic disc at its existing temperature – or as near to it as is possible. Since each magnet might be at a slightly different temperature, or, more likely, since the temperature sensing thermistors are not exactly matched, it is essential to find the temperature set point for each side. With the loop OPEN, adjust both set point potentiometers on the front panel until the corresponding LED becomes dark. Red means the set point is too high, blue that it is too low. At the correct value, the LED goes dark. Figure 2.7 is a simplified block diagram of the controller which shows how this works. The thermistor is in one arm of a Wheatstone bridge and the output of the bridge is compared to the adjustable reference voltage. The difference is amplified by the instrument amplifier. Both the actual bridge signal from the thermistor and the reference voltage are available on BNC connectors on the back panel. You may monitor the thermistor bridge signal to assure yourself that the loop is stabilizing the temperature of the magnets.

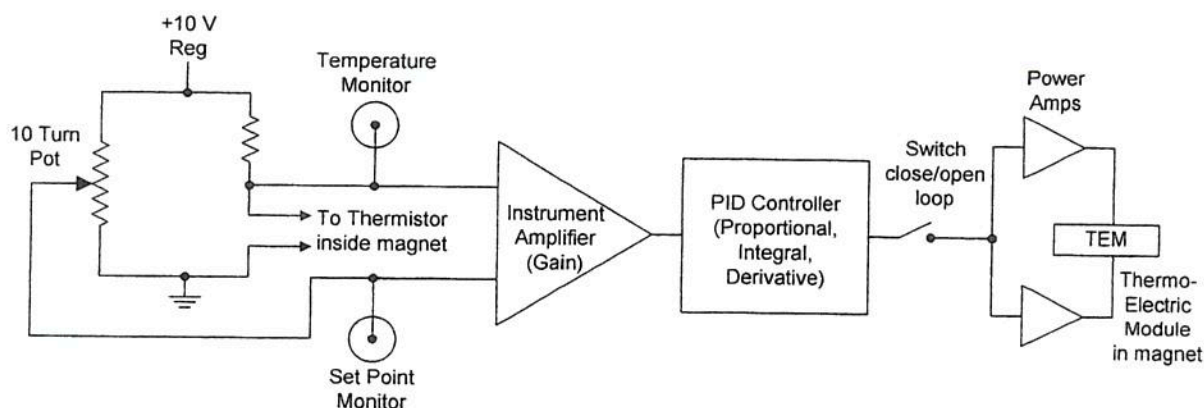


Figure 2.7 Simplified Block Diagram of the Controller

Once the two reference voltages have been carefully set and locked on the ten turn knobs, both loops should be closed simultaneously. You will likely notice some oscillation in the current to the TEC's but this should settle down in a minute or so as the loop PID (Proportions, integral, differential) establishes a fixed temperature. The LED should remain dark as long as the servo is doing its job and keeping the magnets at a fixed temperature. Also the temperature monitor voltage should remain constant and the same as the reference voltage. If this does not occur, repeat the starting procedure. If this system still does not lock the temperature, there may be loose cables or the system may need repair.

Because of the rather slow thermal response times for the entire magnet assembly, it takes about 45 minutes for the magnetic field to achieve optimum stability. Each magnet is slightly different and students may wish to measure this characteristic time as well as the stability of the one they are using. After making sure the field is homogenized and the gradient coils optimized (this can easily be done using a single pulse FID signal with a mineral oil sample), drifts in the magnetic field of as

12. **DO NOT** leave the thermal servo loops on overnight – or on for several days. This will not damage the magnet or the servo electronics, but it may mean that the system will be keeping the magnet at a temperature very far from the current ambient temperature of its surroundings. Part of what makes the PS2-A system so effective at keeping the temperature stable is that the magnet temperature is being locked to the temperature of the local environment at the time the loop was closed. Using a magnet with a servo that has been on too long is not an optimum starting condition for a set of experiments that require maximum field stability.

E.2 Controlling the Temperature

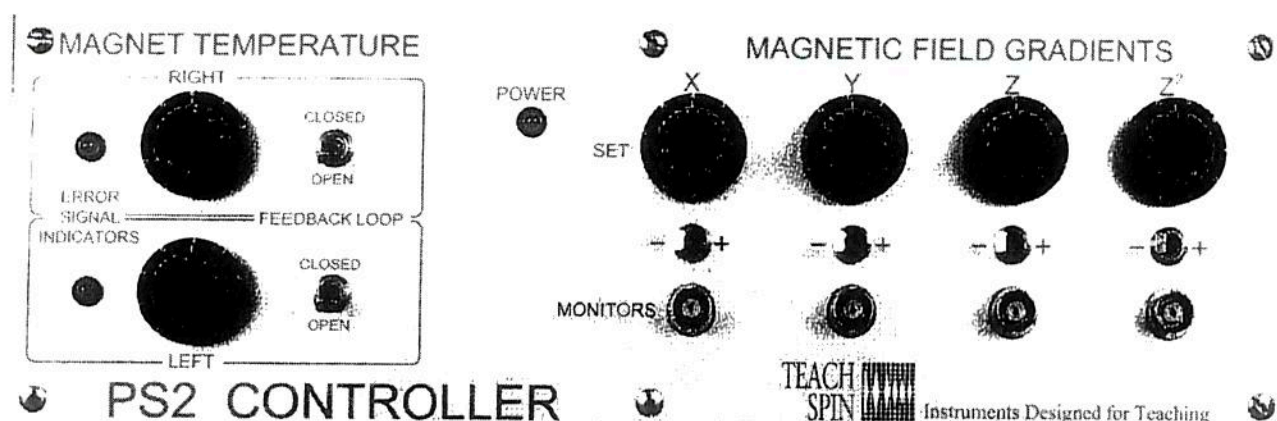


Figure 2.6 Control Panel

The temperature of the NdFeB magnets on the two pole pieces are controlled by two identical and independent thermal servo units, one for each pole. These servo units are designed to keep the temperature of the individual magnetic disc on each pole at a constant temperature. Note that the specific temperature of an individual pole piece is not important. What is essential is that the temperature of each NdFeB disc remains constant for the duration of the experiment.

To accomplish this stability, a special thermistor temperature sensor is put in good thermal contact with each NdFeB disc. A thermistor is a special material whose resistance varies in a predictable and reproducible way with temperature. Thus, the value of the thermistor resistance is a measure of the temperature of the NdFeB disc, which is also a measure of its magnetization. The temperature of the disc is then modulated (controlled) by a thermoelectric cooler, which has the remarkable property of either cooling or heating a surface depending on the polarity of the current through it. Heat is transferred from the NdFeB magnetic to the heat sinks which exchange heat to the room by both convection and radiation. These heat sinks also masquerade as the side panel labels of the unit. Students should not put their hands on these heat sinks during critical runs where optimum temperature stability is required. Some care should be taken with these panels. Letting heavy objects hit them could damage the thermoelectric cooler.

Operating the servo loop temperature controller is straight forward. The last student who used the unit should have left the loop OPEN! This is easily done with the two toggle switches on the front panel. It can also be accomplished by turning the power off, using the power switch on the power entry module at the rear of the Mainframe. Using the power switch, however, is not the best practice because it will also turn off the electric field shim gradient coils. If you are using the shim coils for the experiment, they should have been left ON. The shim coils can be left on indefinitely

are mounted on each side behind the pole tips. Neodymium-Iron-Boron material was selected as the permanent magnet because of its large magnetization and its ability to retain that magnetization. Simply put, the magnet will likely outlast the electronics, the professors, and even the students.

However, NdFeB has one characteristic that can be a problem. Its magnetization is temperature dependent. Thus, the field in the air gap depends on the temperature of the magnetic material. For many experiments, this drifting magnetic field will cause only a small error in the data. For other experiments, the drift is intolerable. In order to provide a wide range of experimental possibilities, TeachSpin designed a complete temperature control system that is easy to use and will regulate the temperature of the NdFeB magnets. The temperature of the magnetic material is kept so stable that the field in the gap remains essentially constant. Within 50 minutes after the regulation loop has been closed, the gap field will reach an impressive stability.

Field Stability: $\pm 5 \times 10^{-4}$ mT over 15 minutes

(In frequency units, for proton resonance, the stability is ± 20 Hz.)

This, or even better, field stability is obtained if and only if the magnet is used in a reasonable laboratory environment.

The following is a list of “Do’s and Don’ts” that must be respected for the optimum performance of this high stability-high homogeneity magnet.

Do:

1. Place the magnet in a reasonably temperature stable location.
2. Place the magnet on a non-ferromagnetic table.
3. Check all electrical connections to the magnet, temperature controller and spectrometer main frame carefully before attempting any experiment.

DO NOT:

1. Place the magnet in a draft or near an open window.
2. Place the magnet near a vent or a radiator heater.
3. Place the magnet near AC high power lines.
4. Place the magnet near a power transformer – particularly the transformer at the right rear of the main frame of the spectrometer. (There is a reason why we always show the instrument with the magnet to the left of the controller!)
5. Place anything under the magnet that might obstruct the flow of air to the thermoelectric heat sinks.
6. Direct a strong light onto the magnet, especially light from an incandescent source.
7. Keep any significant ferromagnetic materials nearby such as a screw driver, a student belt buckle, steel chain . . . Be particularly careful with items that can move.
8. Move the magnet to a new location during or close to the time of starting an experiment, especially if you want to retain the maximum field stability.
9. Place your hand or other objects on the exposed heat sinks of the thermoelectric coolers.
10. Drop the magnet or give it a large mechanical shock
11. Let ferromagnetic “stuff” fall into the magnet gap - paper clips, pins, small tools, etc

D.4c A - Sweep Amplitude

The sweep amplitude menu item selects the amplitude of the sweep range for the **Auto**, **Triggered**, **Manual** and **Rapid Scan** modes. Sweep ranges of ± 1.953 , ± 3.906 , ± 7.812 , ± 15.625 , ± 31.25 , ± 62.5 , ± 125 , and ± 250 mA are supported.

D.4d D - Sweep Duration

The sweep duration menu item selects the duration of the field sweep for the Auto and Triggered modes. Sweep durations of 10, 20, 50, 100, 200 and 500 seconds are allowed.

E. THE PERMANENT MAGNET

E.1 Overview

The PS2-A spectrometer comes with a uniquely designed permanent magnet described in this section. The physical specifications of the magnet are as follows:

Field Strength in Gap: $0.50 \pm .01$ Tesla
 Homogeneity (minimum): 0.5 mT over 0.125 cm^3
 Gap Width: 1.78 cm
 Pole Diameter: 10.1 cm
 Permanent Magnet Material: NdFeB
 Weight: 15 kg



Figure 2.5 The Permanent Magnet

Removing the wooden top and insulation exposes the basic “H” – frame geometry of the magnet assembly. Soft iron is used for the frame members and the NdFeB permanently magnetized disks

can be varied from 0.02 μs to 20.02 μs . Appearing on the menu as tau:, the time, τ , between the A and B pulse can be varied from 0.0001 s to 9.999 s. The repetition time, written on the menu as P, Period: is the time delay between repeats of the **entire pulse sequence**, meaning the complete pattern of the single A pulse and the chosen number of B pulses. The repetition time can range from 0.2 milliseconds to as long as 100 seconds. If a longer repetition time is required, the Man Start button can be used as a manual start of the pulse sequence. Set the menu to Period: manual to use this feature.

The module provides a short ($\sim 1 \mu\text{s}$) trigger pulse to trigger a 'scope or a computer on either the A or the B pulse. The choice of pulse is made with the toggle marked Sync. Toggle switches are provided on both the A and B pulses to allow either pulse to be turned off as desired. The toggle marked MG is used to disable the Meibohm-Gill 90° phase shift between the A and B pulse in a pulse train that measures T_2 with multiple B pulses. The Blanking Out pulse is a pulse slightly longer than A and B which disables the receiver during the pulse and thus prevents internal capacitors from charging during the RF pulse. This facilitates a faster recovery of the receiver.

Finally, the Ext Start-input requires a TTL pulse which will start a pulse string of the one A pulse and the programmed number of B pulses instead of the internal repetition cycle. Thus, the pulse sequence can be synchronized with any data collecting instrument that the user chooses.

C.2 Specifications

1. A – A Pulse(one only) A_len: 0.02 – 20.02 μs
2. B – B Pulse (0-100_ B_len: 0.02 – 20.02 μs
3. τ tau: 0.0001 – 9.999 seconds
4. N Num_B 0 – 100
5. P Period: 0.2 ms to 100 s
6. External Start TTL Pulse 4 volts 1 μs
7. Manual Start Button
8. Sync on either A or B pulse
9. Sync Out – 0.5 μs TTL Pulse

~~D. LOCK-IN, SWEEP MODULE~~

~~D.1 Overview~~

~~The lock-in sweep module is used for continuous wave (CW) experiments. It is comprised of a two channel lock-in amplifier and a sweep ramp generator with a current output amplifier. Depending on the operating mode, one or both of these components may be in use simultaneously. The following sections describe the operation of this module.~~

Pulsed RF Out signal can be adjusted, quasi-continuously (in one degree steps) by the selector knob using P in the menu. This signal can be switched off by the toggle above the BNC connector

2. CW Out: This is a continuous wave sinusoidal signal that is used for CW resonance detection. This signal is fed into the RF probe via the directional coupler inside the receiver module. Its amplitude can be adjusted in this module when the menu is set on A. The power is measured in dBm and can be varied in one dB steps. **This output must be connected to the CW In on the Receiver but the BNC cable must be removed when operating in the pulse mode.** The CW output can be turned off by the toggle above the connector.
3. Pulsed RF Out: This is the high power pulsed RF signal that is sent to the coil surrounding the sample via the receiver module. This is a reverse “gender” BNC connector which requires the reverse gender BNC cable to connect it to the RF Pulse In on the receiver module. The cables have “female” BNC connectors, and the BNC panel connectors are male. It is essential that the correct BNC cable is used. The usual BNC male cable connector will not work and should not be forced on.
4. Pulse In, I and Q: These are both inputs connected to the respective I and Q Pulse Out on the Pulse Programmer module. They receive the timing pulses that shape the RF output power used to “tip” the nuclear spins.
5. Sweep In.: This may be used to sweep the RF frequency with an external analog voltage. It may be used in observing CW NMR resonance. It should not be connected in pulse mode operation.

B.2 Specifications

Frequency: 1 – 30 MHz

Frequency Stability: ± 50 ppm

Frequency Increments: steps of 100 KHz, 1 KHz, 10 Hz

Phase: -180° to $+180^\circ$ in one degree steps

CW Amplitude: -10 dBm to -65 dBm in 1 dB steps

Sweep: 0, 1, 2, 5, 10, 20, KHz/Volt

C. PULSE PROGRAMMER

C.1 Overview

As its title implies, this is the module that produces timed pulse sequences used in the various pulsed NMR experiments. This unit sets the pulse lengths for what are called the A and B pulses. In pulse mode, this unit cannot change the amplitude of the RF pulse, we must adjust the **pulse length** to achieve a 90° or 180° rotation of the magnetization – the so-called “ninety or one hundred eighty degree” pulse.

There is only one A pulse in any pulse sequence. There can be as few as zero and as many as 100 B pulses. The number of B pulses appears on the menu as N, Num_B:. The length of the A pulse appears on the menu as A; A_Len: and the length of the B pulse as A B_len:. Both pulse lengths

quadrature outputs then represent the two orthogonal phases of the signal reflected from the sample coil. A good way to display these signals is to place your oscilloscope in X-Y mode with equal sensitivity settings (about 1V/div) for both the X and Y axis. Set the beam to be centered in the screen when the X and Y inputs are at zero volts. The vector from the center of the screen to the displayed beam position then represents the amplitude (distance from the screen center) and phase (angle from the screen center) of the reflected signal.

There is an unknown phase constant (rotation about the center of the screen) arising from propagation delays from both the cable connecting the sample coil to the receiver and the amplifiers and filters in the receiver module. Both the gain and phase of the reflectometer may be calibrated by replacing the sample coil with a component with a known reflection. The simplest component to use is a resistor with a value slightly different from the nominal 50 Ω impedance of the coaxial cables. The amplitude and phase of the reflected signal may be calculated using equation 2.1. By varying the phase of the **Ref. Out** signal from the synthesizer module (using the **Phase** menu item) you can rotate the display of the reflected signal on the oscilloscope so that the real component lies along the X-axis. A word of caution -- The reflectometer in the PS2 receiver will only operate accurately when the reflection coefficient of the component connected to the Sample Coil connector is relatively small. This limitation arises from the fact the the input of the LNA is not well matched to 50 Ω , so that a standing wave can be set up between the LNA input and the device being measured.

The standing wave is small and has little effect on the measurement when the reflection coefficient of the item being measured is small, but will result in significant errors when the reflection coefficient is large, such as when a short circuit or open ended cable are present. The reason is that the input of the LNA amplifier is not matched to 50 Ω arises from the matching requirements needed to achieve low noise operation. In general the impedance matching requirements needed to achieve low noise operation are different from those needed to achieve a low reflectivity input match to the amplifier.

B. 21 MHz SYNTHESIZER

B.1 Overview

The synthesizer module is the source of the radio frequency (RF) signals used in the spectrometer. It is a modern, digitally synthesized, high stability, signal source with a wide frequency range. This unit can produce RF signals from below 1 MHz to over 30 MHz, but that is a much larger range than is needed for the operation of the NMR spectrometer. The proton and fluorine precession frequencies are only 6% different, but the field of the permanent magnet varies from unit-to-unit by several percent and will also drift with ambient temperature. Your unit has the proton's precession frequency in your magnet, at ambient room temperature, marked on the serial label. The marked frequency is a good place to start your "tuning up" of the spectrometer.

The unit has six (6) BNC connectors, three inputs and three outputs. The function of each of these connectors is described below.

1. REF Out: This refers to "reference" output signal that is connected to the **REF. In** on the receiver module. It provides a continuous wave, sinusoidal signal (CW) to the phase sensitive detectors in the receiver. The phase of this signal, relative to the phase of the

A.2b Operation Pulsed Mode - Between Pulses

The operation of the input section of the receiver between pulses can now be understood. (See Figure 2.2a.) For this discussion we can ignore the presence of the directional coupler in the circuit. Between pulses, no signal is coming from the pulsed RF output of the synthesizer (by definition!), and the induced voltages in the sample coil produced by the precessing nuclear spin magnetization is very small (typically 10's of μV) compared to V_0 . In this case both back-to-back diode pairs act like open circuits, so the pulsed RF output of the synthesizer module is effectively disconnected from the sample coil, and the sample coil is connected to the receiver input amplifier through the $\lambda/4$ length of coaxial cable in the receiver module. (See Figure 2.2b) The back-to-back diodes across the input of the receiver module have no effect on the very small signal entering the input of the receiver.

A.2c Operation Pulsed Mode - During RF pulses

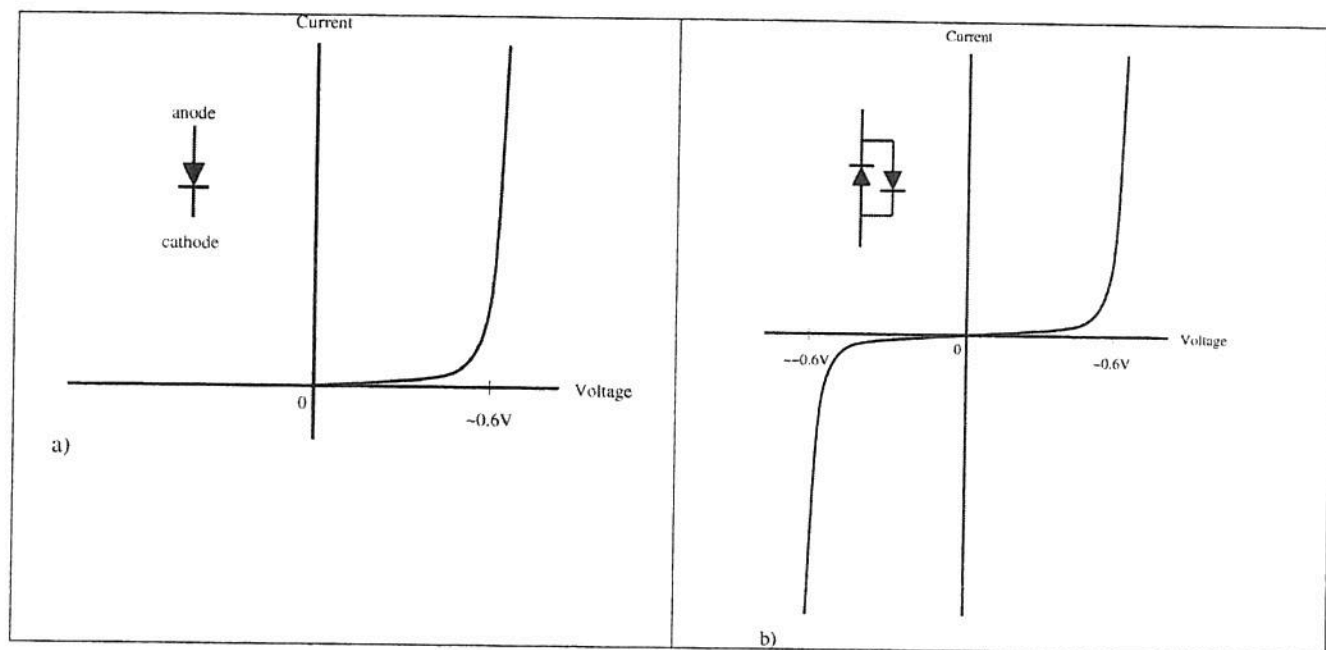
During an RF pulse the pulsed RF output of the synthesizer module swings about $\pm 25\text{ V}$. The diodes connecting the pulsed RF output of the synthesizer to the sample coils may therefore be treated as short circuits (low impedance), so they directly connect the pulsed RF output to the sample coil as desired. (See Figure 2.2a) But wait... there seems to be a problem... the back-to-back diodes across the input of the low noise amplifier also seem to have a large voltage applied to them, and therefore can also be treated as short circuits. If these are in a low impedance state they would appear to be shorting the RF applied to the sample coil to ground through the $\lambda/4$ coaxial line. In the low impedance state, the impedance of the diodes is much less than the 50Ω characteristic impedance of the $\lambda/4$ line, so there is wave reflected from this end of the $\lambda/4$ line 180° out of phase with the wave incident on the diodes. This reflected wave sets up a standing wave on the $\lambda/4$ cable, with a node at the input to the low noise amplifier (point Y), and an antinode at the opposite end of the $\lambda/4$ line (point X). Nodes in the voltage standing wave correspond to low impedance points, while antinodes correspond to high impedance points. Since the end of the $\lambda/4$ cable where the RF pulse is being coupled (point X) from the synthesizer is an antinode, this point is a high impedance, and may therefore be treated as an open circuit! So the back-to-back diodes at the LNA inputs are not shorting the RF pulse to ground, since the $\lambda/4$ length of cable transforms the low impedance at its one end to a high impedance at the other end. The RF pulse from the synthesizer is therefore applied to the sample coil (See Figure 2.2c.) as desired.

A.2d Operation Pulsed Mode - Blanking

Even with the back-to-back diodes and $\lambda/4$ cable, the voltage at the input of the LNA during an RF pulse is many orders of magnitude larger than the voltage induced in the sample coil by the precessing nuclear spin polarization. In order to insure a rapid recovery after the RF pulse ends, *blanking* has been incorporated into the receiver. The **Pulse Programmer** outputs a blanking signal which starts about 10 ns before the RF pulse begins and ends when the RF pulse is complete. The receiver uses this signal to reduce the gain of its RF amplifiers during the RF pulse to prevent the amplifiers from overloading, and to prevent the capacitors in the outputs of the detectors from being charged. The blanking circuitry has the ability to extend the blanking period past the end of the RF through use of the **Blanking Duration** control. This allows for optimizing the blanking interval to compensate for different sample coil tunings.

A.2a Diodes in a Nutshell

For those of you unfamiliar with diodes, a diode is a two terminal device which has very non-linear current - voltage (I-V) characteristics. The diodes used in the receiver module are silicon p-n junctions, and have I-V characteristics as shown in Figure 2.3a.



Figures 2.3 (a) Single diode characteristics

(b) Back-to-back diode characteristics

The current through the diode I , may be roughly described as an exponential function of the voltage applied between the diode's two leads, V

$$I = I_0 e^{\frac{V}{V_0}} \quad (2-1)$$

where V_0 is about 0.6 V, I_0 is a positive constant, and we take V as positive when the diode's anode is more positive than the cathode. For applied voltages much smaller than V_0 very little current flows, so the diode can be treated as a high impedance, or open circuit. (Actually, the diode has a small amount of capacitance, so a slightly more accurate model would be that of a small capacitor.) Conversely, when applied voltages are large compared to V_0 , very large currents flow, so the diode can be modeled as a low impedance, or short circuit. The above discussion works when $V > 0$ (positive voltages). Note that very little current flows for any value of $V < 0$. The RF signals we are dealing with, however, are both positive and negative, so rather than using single diodes, they are used in 'back-to-back' pairs, as shown in Figure 2.3b. In this configuration, when $|V| \ll V_0$ the pair of diodes may be treated as an open circuit, and when $|V| > V_0$ they can be treated as a short circuit.

The input section of the receiver module is designed for use in both continuous wave (CW) and pulsed experiments. It functions slightly differently in these two applications, so they are described separately. To start with we assume that the receiver coil is tuned so that it matches the 50 Ohm impedance of the coaxial cable connecting the sample coil to the **Sample Coil** connector on the receiver module. Chapter III, Sections C.1 and C.3 describe how the sample probe is tuned to satisfy this requirement.

A.2 Pulsed Mode Operation

In pulsed mode operation, the pulsed RF output of the synthesizer module must be coupled to the sample coil to produce the B_1 field that rotates the direction of the nuclear spin polarization. During the RF pulse, the input of the receiver must not interfere with the application of the RF pulse to the sample coil. Once the RF pulse is complete, the input of the receiver must be coupled to the sample coil so that it can amplify the induced RF signal, without interference from the Pulsed RF output of the synthesizer. Essentially, the sample coil should be connected to the synthesizer's Pulsed RF output and disconnected from the receiver during RF pulses. Between RF pulses, the opposite should be true; the sample coil should be connected to the receiver input and disconnected from the synthesizer's Pulsed RF output. See Figure 2. This entire changeover must take place in microseconds, so that signals with short T_2 may be observed. This bit of magic is performed using non-linear devices called diodes and resonant transmission lines.

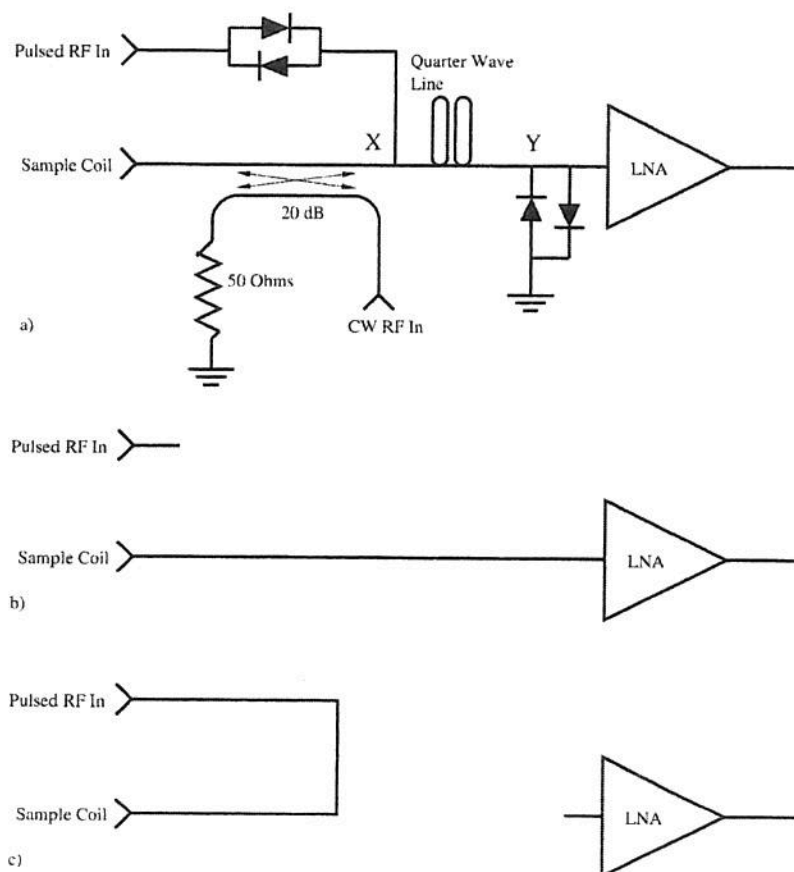


Figure 2.2 (a) PS2 input section circuit. (b) Receiver input section operation, RF pulse off. (c) Receiver input section, RF operation pulse on.

II. THE INSTRUMENT

A. RECEIVER MODULE

A.1 Overview

The basic function of the receiver module is to amplify the small voltage induced in the sample coil by the precessing nuclear spin polarization to a large enough amplitude where it can be easily displayed on an oscilloscope. After passing through the input coupling circuitry, the signal from the sample coil is amplified by a fixed gain low noise amplifier (LNA) with approximately 20 dB of gain and noise figure of approximately 2.5 dB.

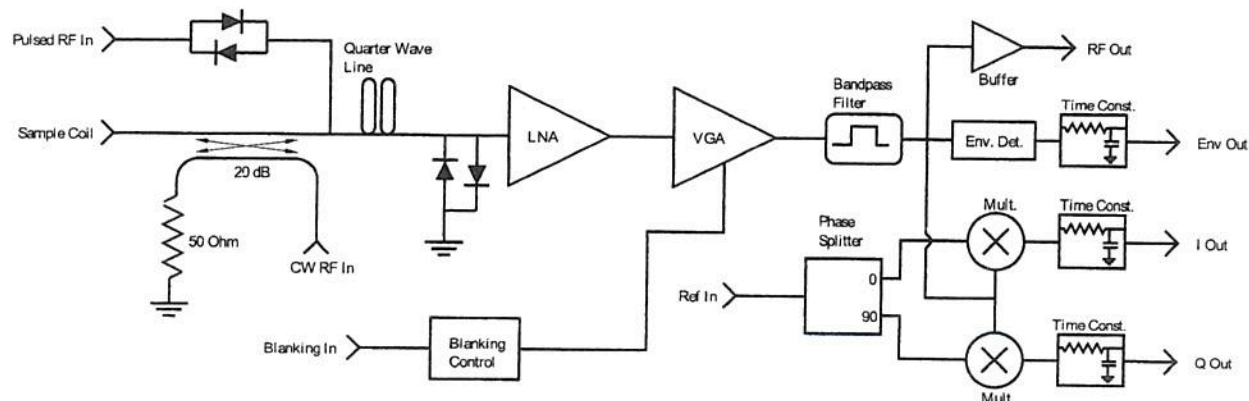


Figure 2.1

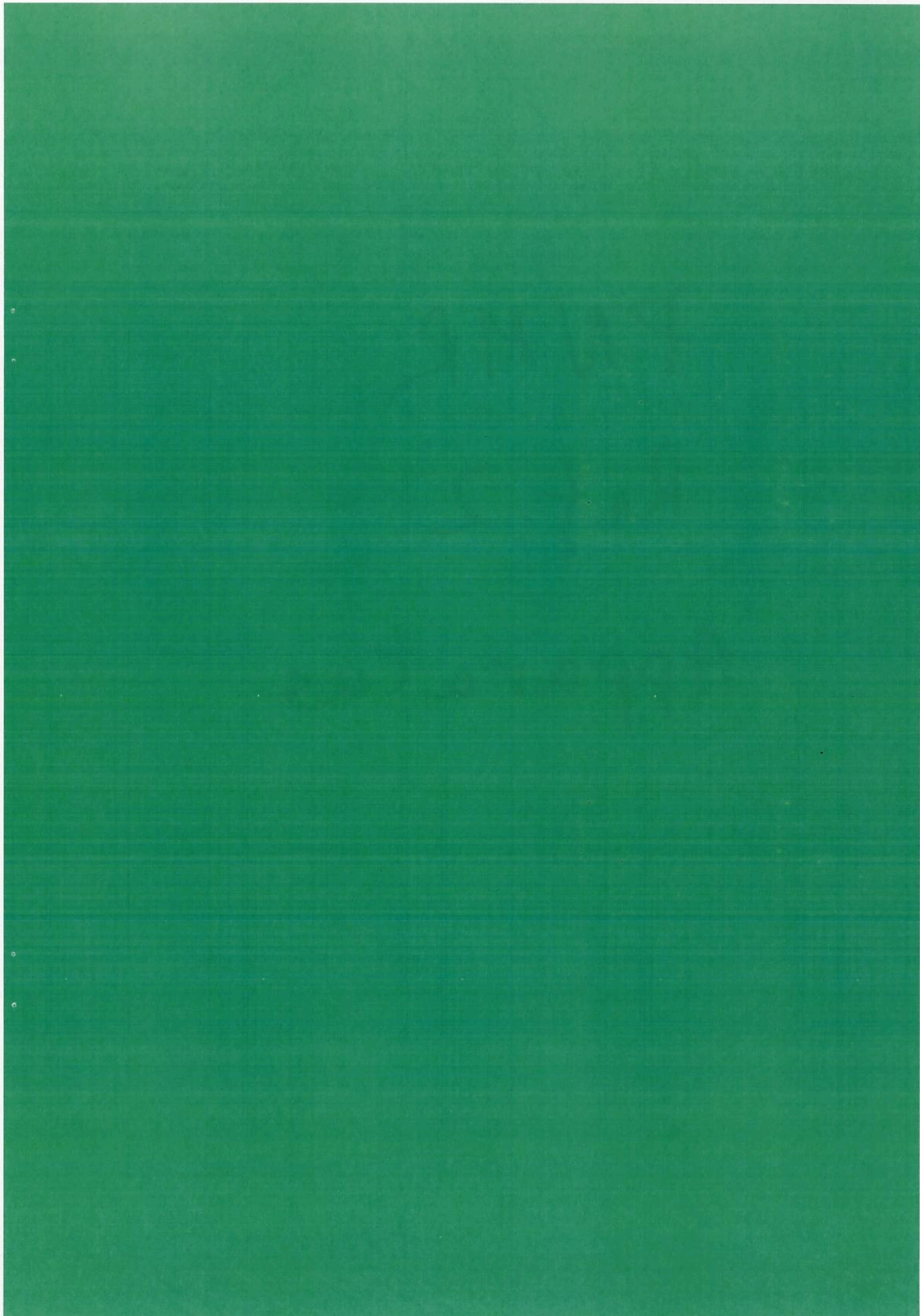
The output of the low noise amplifier is fed into a variable gain amplifier. The gain of this amplifier can be varied from 0 to 80 dB using the **Gain** control. Both of these two amplification stages are designed to have a wide enough bandwidth so that they can amplify the signals from both protons and fluorine nuclei, and have fast recovery times for pulsed NMR experiments. The output of the variable gain stage is then filtered by a narrow band filter to eliminate out-of-band noise. The center frequency of the narrow band filter can be switched between two values, corresponding to the proton and fluorine resonant frequencies, using the **Band** switch marked p and f (for proton and fluorine). A buffered version of the filter output is available at the **RF Out** connector.

The output of the filter is also sent to the inputs of the envelope detector and the phase sensitive detectors. The envelope detector tracks the amplitude of the input RF voltage and drives the **Env. Out** connector on the receiver module. The **I Out** signal from the phase sensitive detector is the product of the **Ref In** signal and the filter output, while the **Q Out** signal is a product of the **Ref In** signal shifted 90° and the filter output. By adjusting the phase of the **RF Out** signal from the Synthesizer module, you can make these signals correspond to the in-phase and quadrature components of the NMR signal. The outputs of all three detectors (**I Out**, **Env. Out**, and **Q Out**) are filtered by single pole RC filters (6dB/octave) as determined by the setting of the **TC** control on the receiver module panel.

PNMR

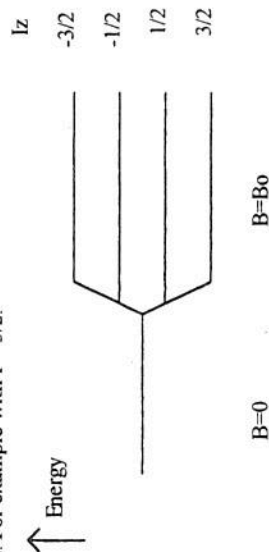
Ref. 2

Apparatus



12. QUANTUM MECHANICAL DESCRIPTION OF MAGNETIC RESONANCE

From the point of view of Magnetic Resonance Imaging (MRI) it is sufficient in general to employ a purely classical description of NMR. However the magnetic resonance phenomenon can of course be described in terms of a quantum mechanical picture. We saw previously that a nucleus in its ground state can be characterised by a spin angular momentum quantum number I which may be integral or half integral and a magnetic moment $\mu = g\hbar I$ where g is the gyromagnetic ratio of the nucleus, $\hbar = h/2\pi$ and $|I| = \sqrt{I(I+1)}$. According to quantum theory the expectation values of I_z are limited to one of the $(2I+1)$ values $m = I, (I-1), \dots, -I$. In the presence of a static magnetic field B_0 , this $(2I+1)$ fold degeneracy of the ground state is lifted, the nucleus acquiring a Zeeman energy $E = -\mu B_0$. For example with $I = 3/2$:



(Similar arguments apply to the case of an atom of a paramagnetic element, of total angular momentum quantum number J and magnetic moment $\mu = -g\beta J$, for which the Zeeman Levels are given by $E = g\beta B_0 m$ where β is the Bohr magneton ($\beta = e\hbar/2mc$) and g the 'g factor').

Magnetic resonance involves the excitation of transitions between the Zeeman levels by application of a suitable perturbation to the spin system.

13. RESONANCE CONDITION AND SELECTION RULES

How can we excite transitions within the Zeeman manifold? According to the Fermi Golden Rule, the transition probability per unit time produced by a time dependant perturbation $V(t)$ is

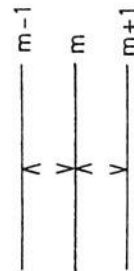
$$W_{mm'} = (2\pi\hbar) |\langle m | V | m' \rangle|^2 \delta(E_{m'} - E_m \pm \hbar\omega) \quad (20)$$

We clearly require a perturbation $V(t)$ which has non-zero matrix elements between states of different m . Consider an oscillating B field $B_1(t) = 2B_1(\cos \omega t) \hat{x}$ (with $B_1 \ll B_0$) for which

$$V = -\mu \cdot B_1 = -\gamma \hbar 2B_1 I_x \cos \omega t$$

We can write $I_x = \frac{1}{2}(I_+ + I_-)$ where $I_{\pm} = I_x \pm iI_y$ and

$$I_{\pm} |m\rangle = \sqrt{I(I+1) - m(m\pm 1)} |m \pm 1\rangle$$



Hence such a perturbation will induce transitions between states for which $\Delta m = \pm 1$ (selection rule) and the off-diagonal matrix elements are non-zero only if $B_1(t)$ is applied perpendicular to B_0 , since only I_x and I_y couple states of different m .

The δ function in equation (1) is non-zero only for

$$\begin{aligned} \Delta E &= |E_{m\pm 1} - E_m| = \pm \hbar\omega_0 \\ &= \pm \gamma \hbar B_0 \end{aligned}$$

Hence we obtain the resonance condition:

$$\omega_0 = \gamma B_0 \quad \text{c.f. equation (1)}$$

Typically for NMR we saw that with $B_0 \approx 1$ T $\nu_0 = \omega_0/2\pi \approx$ MHz, i.e. radiofrequencies (RF) (E.g. for ^1H nuclei, $\nu_0 \approx 42$ MHz in a field of 1 Tesla).

In contrast for ESR, since electronic magnetic moments are $\approx 10^3 \times$ nuclear moments, ESR frequencies are $\approx$ GHz, (i.e. microwave frequencies), in the same field strength.

14. LEVEL POPULATIONS AND SPIN TEMPERATURE

While the resonance condition defines the frequency at which resonance will occur, the magnitude of the NMR signal will depend on the sample magnetisation, itself a function of the populations of the levels between which transitions are excited (see below). In equilibrium the relative populations of adjacent levels are given by the Boltzmann factor:

$$\frac{N(m-1)}{N(m)} = e^{-\Delta E/kT} = e^{-\gamma \hbar B_0/kT}$$

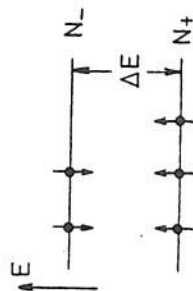
For a spin $1/2$ system, if the total number of spins is $N = N_+ + N_-$, then the population difference $n = N_+ - N_-$ is given by

$$\frac{n}{N} = \frac{N_+ - N_-}{N_+ + N_-} = \frac{1 - e^{-\Delta E/kT}}{1 + e^{-\Delta E/kT}}$$

In practice at room temperature $\Delta E/kT \ll 1$ so that

$$\frac{n}{N} \approx \frac{\Delta E}{2kT}$$

Ex for ^1H nuclei (protons) in a field of 1 Tesla at 300 K, $n \approx 3 \times 10^{-6} N$.



The effect of inducing transitions between adjacent levels will be to equalise the populations of those levels. By symmetry $\langle m|V|m\rangle = \langle m'|V|m'\rangle$ i.e. the upward and downward transition probabilities per unit time induced by the oscillating field, W , are the same (see eqn. 20). Thus

$$\frac{dN_+}{dt} = W(N_- - N_+) = -\frac{dN_+}{dt}$$

or

$$\frac{dn}{dt} = -2Wn$$

Hence if the oscillating field is turned on at $t = 0$ when the population difference $n = n_0$ we have:

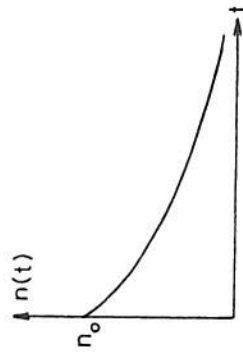
$$n = n_0 e^{-2Wt}$$

ie after some time the population difference is reduced (saturation).

In general for a spin $1/2$ (2 level) system we can define a **spin temperature** T_S such that

$$\frac{N_-}{N_+} = e^{-\Delta E/kT_S}$$

In the absence of a perturbation $T_S \rightarrow T$, the temperature of the surroundings, while in the presence of the oscillating field $T_S \rightarrow \infty$.



15. RELAXATION

The above analysis implies that in the presence of a resonant RF field the spin system will heat up indefinitely. It does not explain how it will return to equilibrium when the RF is turned off, or how the population difference is established when the static field B_0 is turned on. This must result from **spontaneous transitions** with probabilities $W\uparrow$ and $W\downarrow$ corresponding respectively to absorption and emission of energy by the spin system from/to the environment, where $W\downarrow > W\uparrow$. In the absence of an oscillating field then for a spin $1/2$ system

$$\frac{dN_+}{dt} = N_- W\downarrow - N_+ W\uparrow = -\frac{dN_-}{dt}$$

$$\frac{dn}{dt} = 2(N_- W\downarrow - N_+ W\uparrow)$$

$$= (N-n)W\downarrow - (N+n)W\uparrow$$

$$= N(W\downarrow - W\uparrow) - n(W\downarrow + W\uparrow)$$

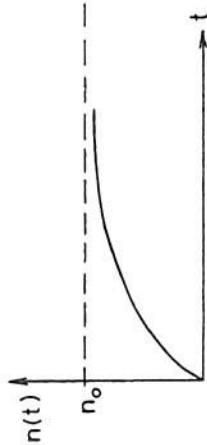
or writing $1/T_1 = W\downarrow + W\uparrow$ and $n_0 = N(W\downarrow - W\uparrow)/(W\downarrow + W\uparrow)$ we have

$$\frac{dn}{dt} = \frac{n_0 - n}{T_1}$$

If $n = 0$ at $t = 0$ this has solution

$$n(t) = n_0(1 - e^{-t/T_1})$$

where n_0 is the equilibrium population difference and T_1 the time constant for establishment of the equilibrium populations.



Clearly for a two level system the presence of a population difference implies that the sample becomes magnetised in the B_0 field (nuclear paramagnetism) with a magnetic moment in the direction of B_0 . Conventionally this is taken as the z direction in magnetic resonance so that $M_z \propto n$ and in the absence of the RF field

$$\frac{dM_z}{dt} = \frac{M_0 - M_z}{T_1} \quad (21)$$

This has solution:

$$M_z(t) = M_0(1 - e^{-t/T_1})$$

Consequently T_1 is known as the longitudinal relaxation time for the spin system. Since it involves an interaction of the spin system with its surroundings, the energy of the spin system is not conserved in the relaxation process. Energy is exchanged between the spins and the thermal degrees of freedom of the sample. For this reason it is also known as the **spin-lattice relaxation time**. (N.B. Spontaneous transitions involving the radiation field are negligible at RF frequencies.)

16. STATIC SUSCEPTIBILITY AND CURIES LAW

For the general case of an $I > 1/2$ spin system the z magnetisation of the sample can be computed since each spin contributes $\mu_z = \gamma \hbar m$, so that

$$M_z = \sum_{m=-I}^{+I} N(m) \gamma \hbar m$$

For a spin system described by spin temperature T_S

$$N(m) = \frac{N}{(2I+1)} \exp \left[\frac{\gamma \hbar B_0 m}{k T_S} \right] \\ \approx \frac{N}{(2I+1)} \left[1 + \frac{\gamma \hbar B_0 m}{k T_S} \right]$$

and

$$M_z = \frac{N \gamma^2 \hbar^2 B_0}{(2I+1) k T_S} \sum m^2$$

$$\sum_{-I}^{+I} m^2 = \frac{1}{3} I(I+1)(2I+1)$$

But

$$M_z = \frac{N \gamma^2 \hbar^2 I(I+1) B_0}{3 k T_S}$$

so that

Hence in equilibrium when $T_S \rightarrow T$ and $M_z \rightarrow M_0$

$$M_0 = \frac{\chi_0}{\mu_0} B_0$$

where the static susceptibility

$$\chi_0 = \frac{m_0 N \gamma^2 \hbar^2 I(I+1)}{3 k T} = \frac{C}{T}$$

- Curie's law.

17. SPATIAL RESOLUTION

How can we obtain images with NMR? To do so we must be able to distinguish protons in different parts of the body/sample. This is achieved by application of magnetic field gradients superimposed on the uniform field B_0 of the magnet.

In conventional NMR spectroscopy we try to make the static magnetic field as homogeneous as possible in order that all spins (of a particular chemical species) resonate at the same frequency, yielding a sharply peaked NMR spectrum. However if we acquire the NMR signal in the presence of a magnetic field gradient $G \equiv (dB_z/dx, dB_z/dy, dB_z/dz)$, the precession frequency of a given nucleus will depend on its position r according to

$$\nu(r) = \gamma B(r) \\ = \gamma (B_0 + G \cdot r) \quad (23)$$

For example if the field gradient is applied in the x direction (G_x), all protons with a particular x coordinate will resonate at a unique frequency, different from that of protons in other parts of the body. Thus the precession frequency of a particular spin, in the presence of a magnetic field gradient, labels its spatial coordinate in the direction of the gradient. The NMR spectrum then becomes a projection of the spin density onto the gradient direction. The overall spread of frequencies across the sample is called the profile width W and is given, for a sample of dimension D in the gradient direction by

$$W = \gamma G_R D \quad (24)$$

where G_R in general is the 'readout' gradient.

18. SLICE SELECTION

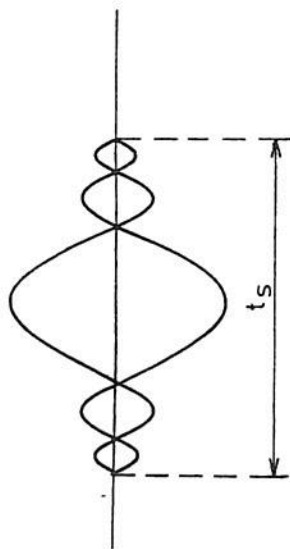
In practice we generally wish to image separate slices through the body. (It is difficult to display a truly three dimensional image!) We therefore select a particular slice to image by applying a specially shaped RF pulse, of well defined (narrow) bandwidth, in the presence of a slice selection gradient G_S . The effect of such a pulse will be to perturb or excite only those nuclear spins whose Larmor precession frequencies lie within the range spanned by the bandwidth of the RF pulse. These will be found in a slice through the body/sample perpendicular to G_S , the thickness of which is determined by the bandwidth of the pulse and the magnitude of the slice selection gradient

$$ds_l = \Delta \nu_s / \gamma G_S \quad (25)$$

The bandwidth of the slice selection pulse $\Delta \nu_s$ depends on the details of its shape. Typically since (i.e. $\sin(\omega t)/(\omega t)$) or gaussian shaped pulses are employed for slice selection, the former yielding a rectangular shaped frequency spectrum and hence a uniform pulse tip angle across the slice. For the case of a 'sinc n' pulse, (a sinc pulse with n lobes), of duration t_S

$$\Delta \nu_s = 2n/t_S$$

Ex Sinc 3



For a more detailed analysis of slice selection, see e.g. Bailes and Bryant, Contemp Phys. 25, 441-475, (1984).

19. FREQUENCY AND PHASE ENCODING

Spatial resolution within the plane of the selected slice can be achieved in a number of different ways. In projection reconstruction or back projection MRI methods, following excitation of the selected slice, the resulting FID signal is recorded in the presence of a readout gradient to yield a projection of the spin density within the slice on to the gradient direction. If the experiment is then repeated with the readout gradient rotated, a set of such projections can be acquired corresponding to different gradient directions, covering a 180° arc. An image of the slice can then be constructed by back projection (or polar 2D (r,θ) Fourier transformation), using algorithms very similar to those employed in X-ray C.T. This method was used by Lauterbur to produce the first NMR image in 1972. He called it 'zeugmatography'.

A more common approach in MRI is to use the frequency of the signal to encode in-plane spatial position in one direction (say x) only. Spatial location in the orthogonal direction (y) is then encoded by applying a stepped magnetic field gradient at right angles to both readout (x) and slice select (z) gradients. The effect of this 'phase encoding' gradient on the nuclear spins is to impart to the transverse magnetisation a phase modulation whose frequency is a function of position in the direction of the phase encoding gradient. By recording a signal (FID or echo) for each value of the phase encoding gradient G_p and stepping the gradient through an appropriate range of values, a 2-dimensional data set is acquired. In principle either the magnitude or duration of the phase encoding gradient may be varied, although in practice the former method is usually employed, in which case the method is also known as 'spin-warp' imaging.

Clearly the frequency of oscillation of the phase angle φ with increasing magnitude of the stepped gradient G_p will be proportional to the contribution G_p makes to the total field at a particular point. Thus the matrix of points S(k_x,k_y) obtained by recording say 128 FID signals, each of which is digitised into 256 data points, will be subject to two 'spatial frequencies'

$$k_x = \int_0^{t_x} \gamma G_R dt = \gamma G_R t_x \quad (26)$$

where t_x is the time from the start of the FID signal or echo peak and

Fig. 2. Schematic spin density projection profile for an inhomogeneous sample comprising an annulus of high spin density (e.g. soft tissue) surrounding a region containing a low density of mobile spins (e.g. bone).

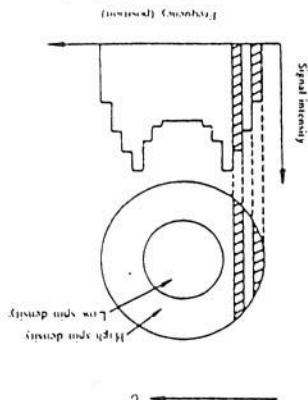
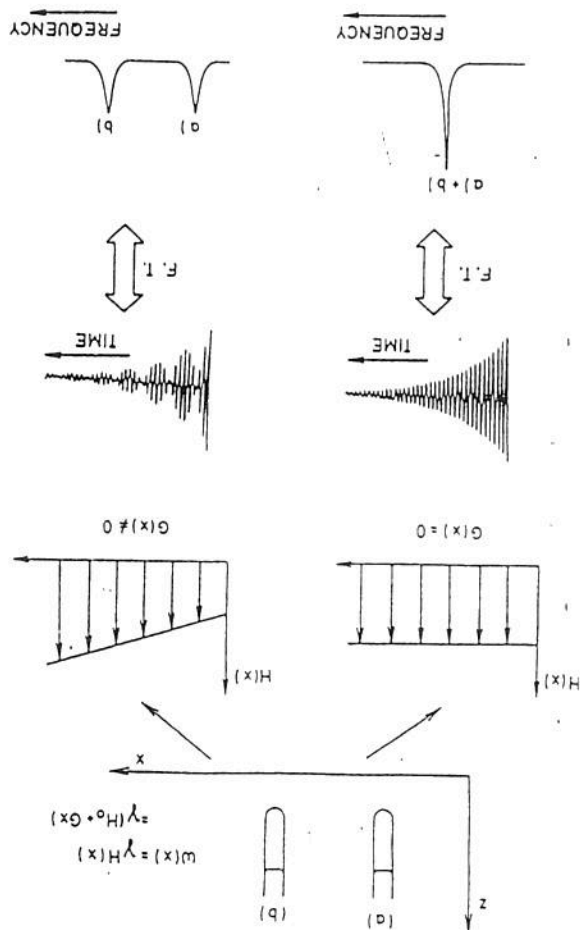
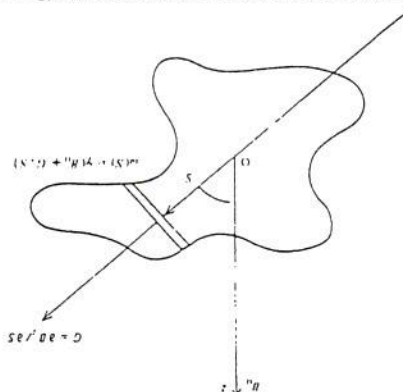
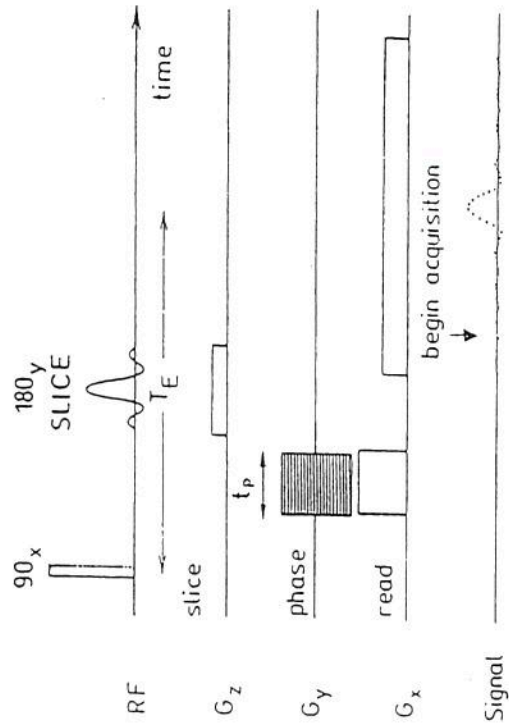
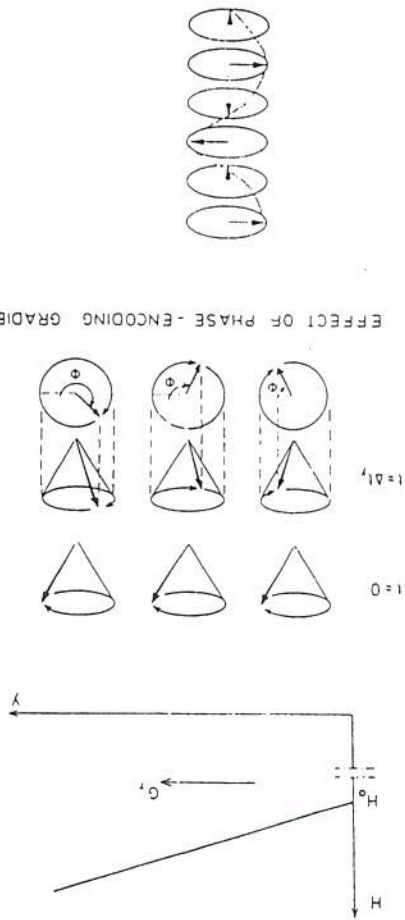


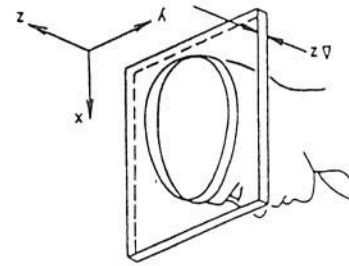
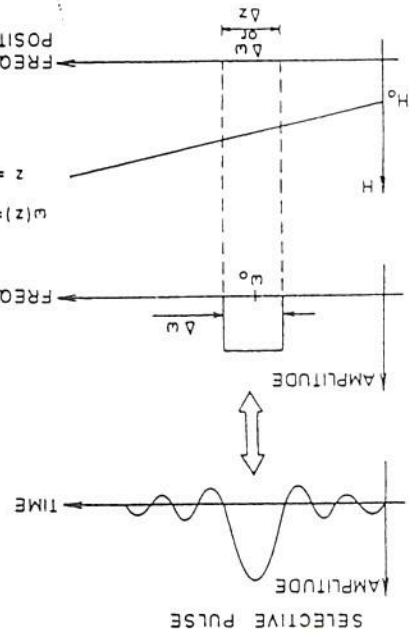
Fig. 1. In the presence of a magnetic field gradient G_z nuclei in different slices of the sample, perpendicular to the gradient direction, precess at different frequencies.



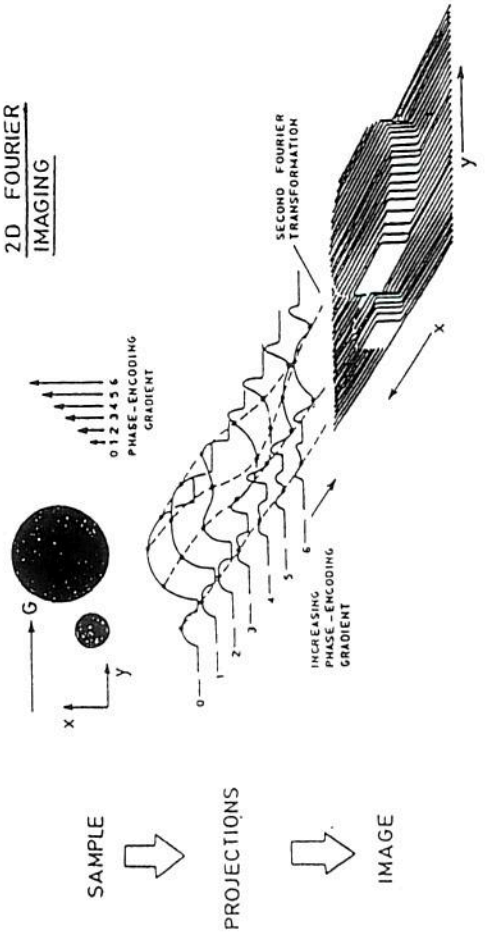
EFFECT OF PHASE-ENCODING GRADIENT



SPIN-ECHO IMAGING SEQUENCE



2D FOURIER IMAGING



$$k_y = \gamma \int_0^{t_p} G_p(t) dt \quad (27)$$

where t_p is the duration of the phase encoding pulse.

The spatial distribution of signal intensity $S(x,y)$, i.e. the NMR image, is then obtained by a two dimensional Cartesian (x,y) Fourier transformation of the signal data

$$S(x,y) = \iint S(k_x k_y) e^{-2\pi i(k_x x + k_y y)} dk_x dk_y \quad (28)$$

For further details see e.g. Callaghan and Eccles, *J. Mag. Res.* 71, 426-445 (1987); Bailes and Bryant, *Contempt. Phys.* 25, 441-475 (1984).

2.0. FIELD OF VIEW AND PIXEL SPACING

In practice the FID or echo signal is sampled at discrete times separated by the dwell time DW which is just the inverse of the sampling frequency F_s . In the case of quadrature detection each phase is sampled at intervals 2DW so that the corresponding spectral width $SW = 1/2DW$ and the effective field of view in the frequency encoding direction

$$\Delta x = SW/\gamma G_R = 1/\gamma G_R (2 DW) \quad (29)$$

If the signal is digitised into N_R real and N_R imaginary points, then clearly the real image will contain N_R pixels in the frequency direction with spacing

$$\delta x = \Delta x/N_R = 1/\gamma G_R N_R (2 DW) \quad (30)$$

But the total signal acquisition time $T_S = (2 DW) N_R$ so that

$$\delta x = 1/\gamma G_R T_S \quad (31)$$

In the phase encoding direction, the phase encoding gradient is stepped over a range of values from $-G_p$ to $+G_p$ in steps of ΔG_p , giving $N_p = 2G_p/\Delta G_p$ steps. The corresponding expression for the pixel spacing in the phase-encoding direction is therefore

$$\delta y = 1/(\gamma G_p t_p) = 2/(\gamma \Delta G_p N_p t_p) \quad (32)$$

and the corresponding field of view

$$\Delta y = N_p \delta y = 2/(\gamma \Delta G_p t_p) \quad (33)$$

The total imaging time is determined by the number of phase encoding steps N_p , the sequence repetition time T_R and the number of samples excitations per phase encoding gradient value N_S .

$$T_I = N_S N_p T_R \quad (34)$$

Typically $N_p = 128$ or 256 and $N_S = 1-4$, i.e. at most 4 samples are acquired for signal averaging to improve the signal to noise ratio (S/N) in clinical MRI. Sequence repetition times range from a few hundred milliseconds to several seconds, yielding imaging times of a few minutes. It should be noted that these imaging times are long compared to those encountered for ultrasound or X-ray C.T. While rapid scan methods of imaging have been developed, reductions in overall imaging time can only be achieved at the expense of reduced spatial resolution. Thus there are still problems in applying MRI to regions affected by irregular motion, such as the upper abdomen.

2.1. CONTRAST IN NMR IMAGES

MRI can, to a great extent be considered as an interleaving between the traditional methods of pulsed NMR, which prepare the nuclear magnetisation in a particular manner at a given time, and the application of magnetic field gradients which, as we will see later, provide the spatial discrimination necessary to generate an image.

The manner in which the sample magnetisation is prepared affects the signal contrast in the resulting image and is determined by the RF pulse sequence employed for imaging. We will consider in detail only a few of the pulse sequences which have been employed in imaging - notably those which have most extensive application in clinical MRI.

In general, the basic sequences are similar to those described previously for relaxation measurements. However for imaging applications, as we have seen, it is necessary to repeat the sequence N_p times, (typically 128 or 256), in order to increment the phase-encode gradient and acquire the information necessary to build up an image. In addition, whereas in relaxation time measurements it is customary to allow the magnetisation to fully recover to equilibrium before application of each sequence, in MRI this is not generally the case. Typically the sequence is repeated with a recycle time $T_R \leq T_1$ in order to (i) acquire the image data more rapidly and (ii) to manipulate the contrast in the resulting image, (see below).

Spin-echo imaging

In the spin echo sequence we saw that a 90° pulse, which interrogates the z magnetisation, is followed after an interval τ by a 180° pulse. This has the effect of reversing the dephasing due to magnetic field inhomogeneity (at least for the case of static spins), so that the magnetisation re-phases to form a 'spin echo' centred at time $TE = 2\tau$ after the 90° pulse. The amplitude of the echo signal is given by:

$$M_y'(0) = M_z e^{-TE/T_2}$$

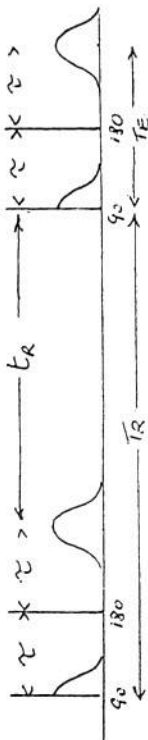
i.e. the acquired signal as a function of time from the echo peak

$$M_y'(t) = M_z e^{-TE/T_2} e^{-t/T_2^*}$$

where here

$$M_z = M_0 [1 - 2 e^{-(T_R - TE/2)/T_1} + e^{-T_R/T_1}]$$

Note that T_R here is defined as the time between successive 90° pulses in the (repeated) spin echo sequence and that the amplitude of the echo depends on TE and T_2 as well as T_R , T_1 and M_0 .



Clearly if $T_E \ll T_R$ then the acquired echo amplitude:

$$M_y \approx M_0 [1 - e^{-T_R/T_1}] e^{-T_E/T_2}$$

If required, further 180° pulses can be added at times $TE_1, TE_2 > TE_1, TE_3 > TE_2$ etc. to generate a train of spin echoes which decay away with time constant T_2 . In this case, the image derived from the first echo, ($TE \leq T_2$), will exhibit contrast determined largely by T_1 and/or spin density variations between different parts of an inhomogeneous sample, ('Partial Saturation' contrast). Those derived from later echoes will exhibit increasing contributions to image contrast from T_2 variations across the sample, (' T_2 weighting').

N.B. the decay of the amplitudes of successive echoes is determined by the **intrinsic** transverse relaxation time of the sample, at least for stationary spins, and is not affected by B_0 field inhomogeneity.

Inversion-recovery imaging

This is a two-pulse sequence used to introduce even stronger T_1 dependence in the image. The spin system is first prepared by applying a 180° which inverts the magnetisation, but leaves it along $-z$ (i.e. antiparallel to B_0) so no signal ensues. The signal is then sampled after a delay T_1 by applying a 90° 'readout' pulse.

Following the 90° readout pulse at time T_1 the z magnetisation starts from zero (assuming $T_1 \gg T_2^*$) and recovers according to

$$M_z = M_0 [1 - e^{-(T_R - T_1)/T_1}]$$

This magnetisation is inverted by the 180° pulse and then continues to relax towards its equilibrium value $+M_0$ according to

$$\begin{aligned} M_z &= -M_0 [1 - e^{-(T_R - T_1)/T_1}] e^{-T_1/T_1} + M_0 (1 - e^{-T_1/T_1}) \\ &= M_0 [1 - 2e^{-T_1/T_1} + e^{-T_R/T_1}] \end{aligned}$$

which for $T_R \gg T_1$ reduces to (c.f. eqn. (16)):

$$M_z = M_0 [1 - 2e^{-T_1/T_1}]$$

Here the acquired signal as a function of time:

$$M_y(t) = M_z e^{-t/T_2}$$

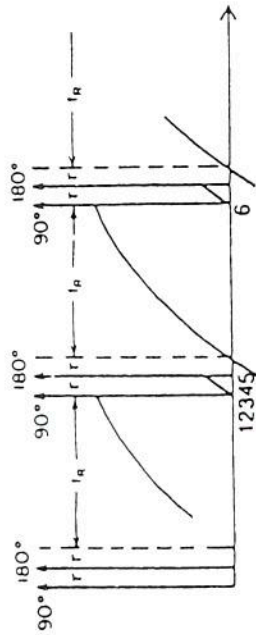
with M_z as above or, if a further 180° pulse is added after the 90° 'readout' pulse, to refocus the signal to form a spin-echo:

$$M_z = M_0 [1 - 2e^{-T_1/T_1} + 2e^{-(T_R - T_E/2)/T_1} - e^{-T_R/T_1}]$$

with:

$$M_y(t) = M_z e^{-t/T_2} e^{-t/T_2}$$

Signal amplitude in spin echo imaging



	M_z	M_y
1. Partially recovered magnetisation	M_z	0
2. Following 90° pulse	0	M_z
3. Relaxation during $\tau = T_E/2$	$M_0 [1 - e^{-\tau/T_1}]$	$M_z e^{-\tau/T_2}$
4. Inversion by 180° pulse	$-M_0 [1 - e^{-\tau/T_1}]$	$-M_z e^{-\tau/T_2}$
5. Relaxation during $\tau = T_E/2$	$M_0 [1 - 2e^{-\tau/T_1} + e^{-2\tau/T_1}]$	$-M_z e^{-2\tau/T_2}$
6. Recovery during $\tau + t_R$	$M_0 [1 - 2e^{-(\tau+t_R)/T_1} + e^{-(2\tau+t_R)/T_1}]$ $= M_0 [1 - 2e^{-(T_R+T_E/2)/T_1} + e^{-T_R/T_1}]$ (since $T_E = 2\tau$ and $T_R = t_R + 2\tau$)	0

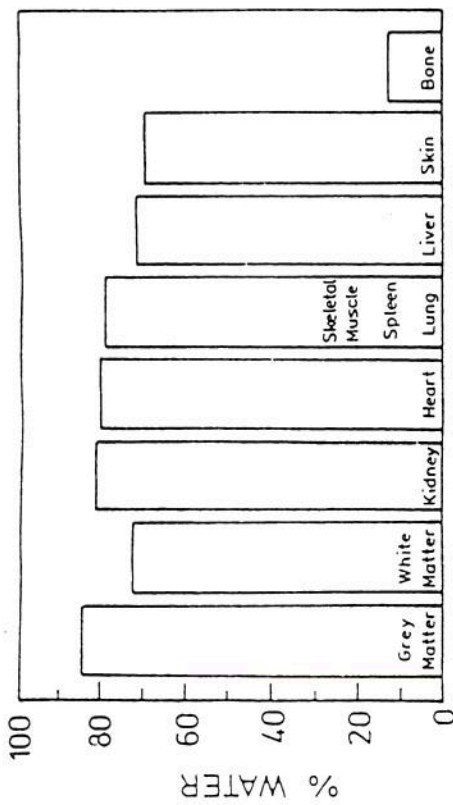


FIGURE 4. — Water content of various body tissues.

22. SIGNAL TO NOISE RATIO IN MRI

A detailed treatment of the signal to noise ratio in MRI is beyond the scope of this review. Detailed discussions can be found in a number of references [see e.g. Edelstein et al, *Mag. Res. in Med.* 3, 604-618 (1986); Callaghan and Eccles, *J. Mag. Res.* 71, 426-445 (1987); Wehrli et al in 'Magnetic Resonance Imaging' Ed. C.L. Partain, Saunders (1987)]. The latter quote for the voxel S/N

$$S/N \propto \frac{U_V \sqrt{N_V} \sqrt{N_P} \sqrt{N_S}}{\sqrt{BW}} \cdot \frac{\omega_0^2}{\sqrt{a R_{Q\phi_0}^2/2 + b R_{P\phi_0}^2}} \cdot N_H(r) R(r, t) \quad (35)$$

where $U_V = \delta \omega \gamma dV$ is the voxel volume, BW is the receiver bandwidth, R_Q and R_P are the resistance of the RF detector coil and patient respectively, N_H is the proton spin density and a, b are geometric constants.

An important feature of MRI is that the patient, being conductive, loads the RF coil, reducing its effective Q and injecting noise into the pre-amplifier. Commonly this term dominates over the Johnson noise due to the resistance of the coil itself. Note that if this is the case

$$S/N \propto \omega_0 \propto B_0$$

In practice the S/N ratio does increase approximately linearly with field strength.

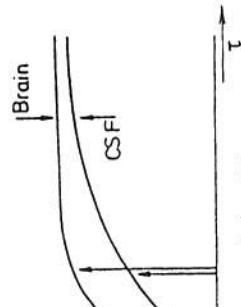
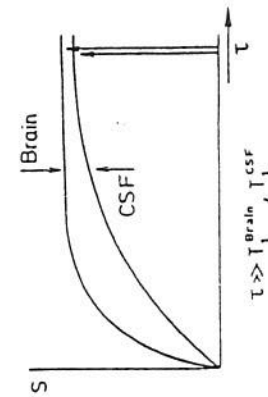
23. IMAGING HARDWARE

The major components of an MRI system comprise:

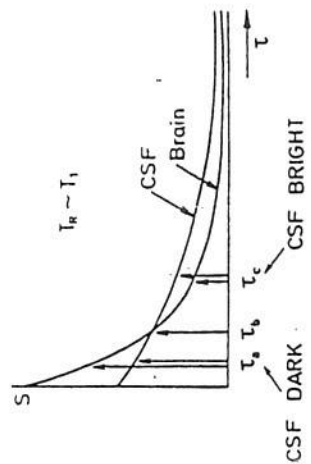
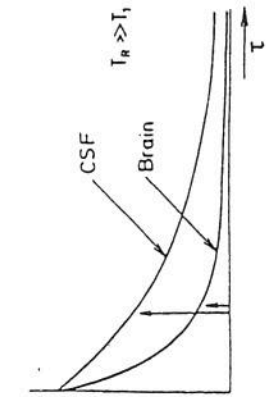
1. Magnet System (and Power Supply).
2. Shim Coils and Power Supplies
3. Gradient Coils and Power Supplies
4. R.F. Coils
5. Frequency Synthesiser
6. Computer and Hard Disk System
7. Sequence Controller
8. RF Modulator
9. RF Transmitter
10. Pre-amplifier
11. Quadrature Phase Sensitive Detector
12. Image Display System
13. Magnetic Tape or Optical Disk for Archiving
14. Multi-format Camera
15. Patient Table and Alignment System.

CONTRAST IN MRI

a) PARTIAL SATURATION (PS)



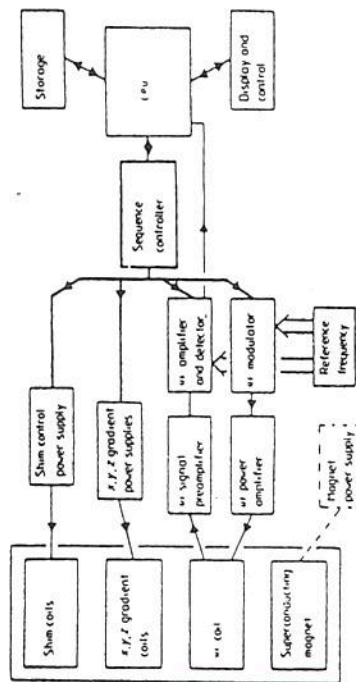
b) SPIN-ECHO (SE)



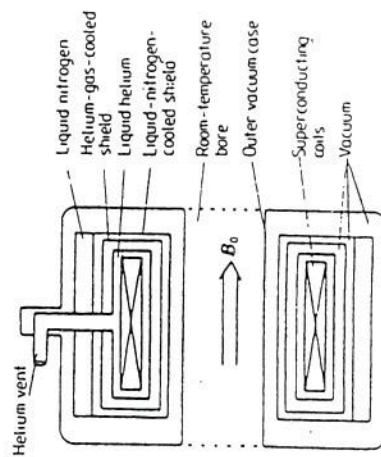
Imaging systems have been constructed with static magnetic fields ranging from 20 mT (200 gauss) to 4.4 Tesla, with most systems having fields in the range 0.3-1.5 T. For fields of 0.3 T and below resistive magnets may be employed, while for fields above 0.3 T, superconducting magnets must be used. Superconducting magnets require cryogenics (liquid nitrogen and liquid helium) to maintain the magnet windings at 4.2K. This makes them relatively expensive to install and operate, but they provide magnetic fields of high stability and the high flux densities achievable result in good signal to noise. Resistive systems are cheaper to install, but require a continuous supply of power, dissipate significant amounts of heat and generally result in inferior image quality. Low field MRI systems have also been based around permanent magnets, but these are limited to field strengths below 0.2 T and weigh up to 3.5 tonnes (compared to 6-8 tonnes for typical air cored resistive or superconducting magnets).

A typical superconducting MRI magnet has a room temperature bore of approximately 1 metre diameter. It must be fitted with resistive shim coils, [to improve magnetic field (B_0) uniformity], gradient coils and a whole body RF coil, bringing the overall diameter available for patient access down to around 60 cm. Gradient coils are typically of the Maxwell pair (G_z) or Golay (G_x, G_y) geometry, since their axis must be parallel to the B_0 field. Magnetic field homogeneity required for MRI is commonly around 20 ppm over a 50 cm diameter sphere at the magnet isocentre. Magnetic field gradients employed are typically ≈ 1 gauss/cm (10 mT/m). RF coils must generate a B_1 field which is transverse to the magnet bore and hence B_0 . Consequently Helmholtz pair or Birdcage geometries are commonly employed. Separate, smaller RF coils are used for headscans and for 'extremities' (knees, ankles, etc.), in order to increase 'filling factor'. All these RF coils are employed both to transmit the RF pulses and detect the NMR signals. They are incorporated in a high Q tuned circuit in order to maximise signal to noise at the Larmor frequency. For high resolution imaging of localised regions (spine, orbit of the eye, inner ear, shoulder, etc.) custom designed 'surface coils' may be employed. These are used as detector coils only, the body coil being used to transmit the RF pulses. Because of the coupling between the transmitter coil and the surface coil, however, the latter must be de-tuned during transmission of the RF pulses to prevent large currents being induced in them during the RF pulse.

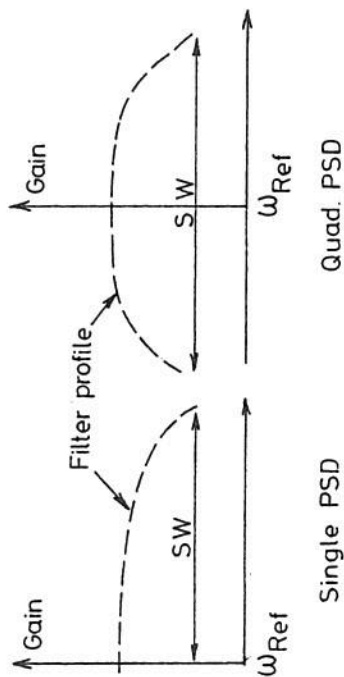
The RF system comprises, in addition to the RF coil, pre-amplifier and receiver system, an RF transmitter capable of delivering both hard and soft pulses. The latter must be linear for pulse shaping. Typically amplifiers capable of delivering around 10 kW of RF power are required, albeit at a low pulse duty factor. In some systems a quadrature transmitter may be used to generate a rotating RF field, thus eliminating the counter rotating component of B_1 which, while virtually irrelevant to signal generation results in a doubling of RF power deposited in the patient. Since power deposition is proportional to ω_0^2 (or B_0^2), this can become a problem in high-field systems (see below). The RF receiver likewise has quadrature phase sensitive detectors, in this case to permit spins with negative precession frequencies (with respect to the reference frequency) to be distinguished from those with positive frequencies. This permits the reference frequency to be placed in the centre of the spectral range, which in turn means that the bandwidth of the receiver filters can be halved, giving a $\sqrt{2}$ improvement in S/N. The receiver filters prevent aliasing of noise from outside the bandwidth of interest. The latter is given by the Nyquist frequency 1/2DW and is typically < 100 kHz.



Block diagram of the major components of an NMR imaging and spectroscopy system.



A diagram showing a cross section through a whole-body superconducting magnet.



As the MRI system is generally based around and controlled by a powerful mini-computer, which facilitates communication with the operator. Imaging sequences may be downloaded to a micro-processor which controls shimming, application of the RF and gradient pulses and signal acquisition via A/D converters. The signals are Fourier transformed using an array processor and passed to an image processor for display. The computer also facilitates imaging storage and archiving.

Careful consideration must be given to the siting of an MRI system and its layout, usually the magnet is contained in a shielded room incorporating a Faraday cage to screen out RF interference. It may also be necessary to provide magnetic shielding so that the stray magnetic field does not extend into areas where it can adversely affect other equipment (Nuclear gamma cameras, image intensifiers, electron beam devices, CT scanners etc.) In addition it should be well removed from large mobile ferromagnetic objects such as trains, lorries and lifts. While the effects of static objects such as structural steel can usually be shimmed out on site, moving objects can adversely affect B_0 homogeneity.

2.4. SAFETY CONSIDERATIONS

Since it does not involve the use of ionising radiations MRI, like ultrasound, is generally regarded as a very safe organ imaging method. The technique involves exposure to the static magnetic field B_0 , time dependent fields arising from the gradients and the RF field B_1 . There is a considerable body of data relating to human exposure to static magnetic fields up to 2T. No evidence of any harmful long term or short term effects has been reported. Exposure to rapidly changing magnetic fields can result in stimulation of the optic nerve giving rise to the appearance of light flashes (optical phosphores). However these do not appear to result in any lasting effects. US authorities have set safety guidelines which limit the rate of change of magnetic fields to 20 T s^{-1} for changes exceeding 10 ms duration while for durations τ_g less than 10 ms they specify

$$\left(\frac{dB}{dt}\right)^2 \tau_g < 4$$

with dB/dt in Tesla/sec and τ_g in seconds. In practice optical phosphores are not reported by patients undergoing routine MR examinations involving standard imaging sequences.

RF deposition in the body can become a problem if it results in significant local or general heating above the level at which the body's circulatory and temperature regulation systems are incapable of removing the additional heat generated. Consequently guidelines on RF power deposition have been set which limit the absorption rate to 4 W/kg in any localised mass of tissue or 0.4 W/kg averaged over the whole body. This ensures that any rise in core body temperature will not exceed 1°C . In practice attempts to measure changes in body temperature associated with routine MR scans have generally resulted in negative results.

There are significant hazards associated with effects of the static field on moveable ferromagnetic objects such as hand tools, steel gas cylinders and trolleys, electronic equipment containing transformers, etc. These can experience a very strong attraction in the non-uniform fringing fields of the magnet and may become dangerous projectiles if handled carelessly. In generally such items should be excluded from the magnet room or bolted down.

Patients with implanted metallic prostheses can pose a particular hazard, particularly if these are ferromagnetic. Consequently patients with heart pacemakers, 'bionic' ear implants and ferromagnetic surgical clips (especially recently inserted aneurism clips) should be de-barred from MR examination and excluded from the magnet area altogether. Non-ferromagnetic prostheses (such as titanium and certain types of stainless steel) will distort the image in their vicinity and may result in some local heating. In general all patients recommended for MR examination (and staff working in an MR unit) should be screened for the possible presence of prostheses as well as metal fragments and shrapnel in sensitive regions.

APPENDIX: GENERAL TREATMENT

In a uniform magnetic field B_0 , the signal following a single 90° pulse was given by an expression of the form, (c.f. eqn 10):

$$S(t) = A Q_{00} \int [B_1(r)/I] M_T(r,0) e^{-i\omega_0 t} e^{i\omega_0 t} d^3r$$

where here we have allowed for a spatial variation of the RF coil sensitivity (corresponding to a non uniform RF field B_1) as well as the transverse magnetisation $M_T(r)$ and relaxation time $T_2(r)$.

More generally in the presence of magnetic field gradients the Larmor frequency $\omega = \omega_0 + \gamma G \cdot r$ so that the signal following some general pulse sequence can be written [see King and Moran Med. Phys. 11, 1-17, (1983)]:

$$S(t) = A Q_{00} \int [B_1(r)/I] M_T(r,0) R(r,t) e^{i\gamma G \cdot r} d^3r$$

where $R(r,t)$ is a function which includes the effects of relaxation as well as the time dependence of the signal due to precession at ω_0 . The latter is removed by phase sensitive detection 'on resonance'. Thus if we define $k = \gamma G t$, we obtain

$$S(k) = \int S(r,t) e^{i 2\pi k \cdot r} d^3r$$

where the 'image'

$$S(r,t) = A Q_{00} [B_1(r)/I] R(r,t) M_T(r,0)$$

contains the explicit time dependence due to relaxation effects only. Clearly then the image can be obtained by Fourier transformation of the acquired signal data according to

$$S(r,t) = \int S(k) e^{-i 2\pi k \cdot r} d^3k$$

The relaxation function $R(r,t)$ depends on the pulse sequence parameters T_E , T_R , T_1 etc. as well as the intrinsic relaxation times T_1 and T_2 which will themselves vary from voxel to voxel. For example in the case of the spin-echo sequence in the limit $T_R \gg T_E$

$$R(r,t) = [1 - e^{-T_E/T_1}] e^{-T_E/T_2}$$

while for inversion recovery

$$R(r,t) = [1 - 2e^{-T_1/T_1} + e^{-(T_1+T_R)/T_1}]$$

Note we have assumed that the B_0 field is uniform in the absence of field gradients so that $T_2^* \rightarrow T_2$.

SAMPLING k SPACE

The concept of k space is useful and instructive in analysing and comparing different approaches to imaging, of which there are now many. Methods can be categorized according to the manner in which k space is sampled. For the case of Fourier or 'Spin-Warp' imaging already discussed, k space is sampled on a square or rectangular raster by stepping the phase-encoding gradient and hence k_y through a series of equally spaced steps and sampling the FID or echo signal at equally spaced times, corresponding to equally spaced values of k_x . Thus the acquired signal

$$S(k) \equiv S(k_x, k_y) = \int_{-x}^{+x} \int_{-y}^{+y} S(x, y) e^{2\pi i (k_x x + k_y y)} dx dy$$

and the image $S(x, y)$ is obtained by Cartesian Fourier transformation of $S(k_x, k_y)$ as indicated previously. Note we will ignore relaxation effects for the time being by setting $R(r, t) = 1$.

In the case of Back Projection imaging in contrast

$$S(k) \equiv S(k, \theta) = \int_0^{\pi} \int_{-R}^R S(r, \theta) e^{2\pi i k \cdot r} r dr d\theta$$

and

$$S(r, \theta) = \int_0^{+k_{max}} \int_{-k_{max}} S(k, \theta) e^{-2\pi i k \cdot r} k dk d\theta$$

Thus each FID $S(k, \theta)$ must be multiplied by $|k|$ and Fourier transformed using a 1-dimensional FFT before back projection to yield the image. Alternatively a 2-dimensional window function may be applied prior to a 2-dimensional Fourier transform to obtain the real space image.

In general to construct an image we first sample $S(k_x, k_y)$ or $S(k, \theta)$ over all four quadrants in reciprocal space. For the case of spin echo imaging we take the peak of the echo as our time zero, corresponding to $k_x = 0$ since at the echo peak

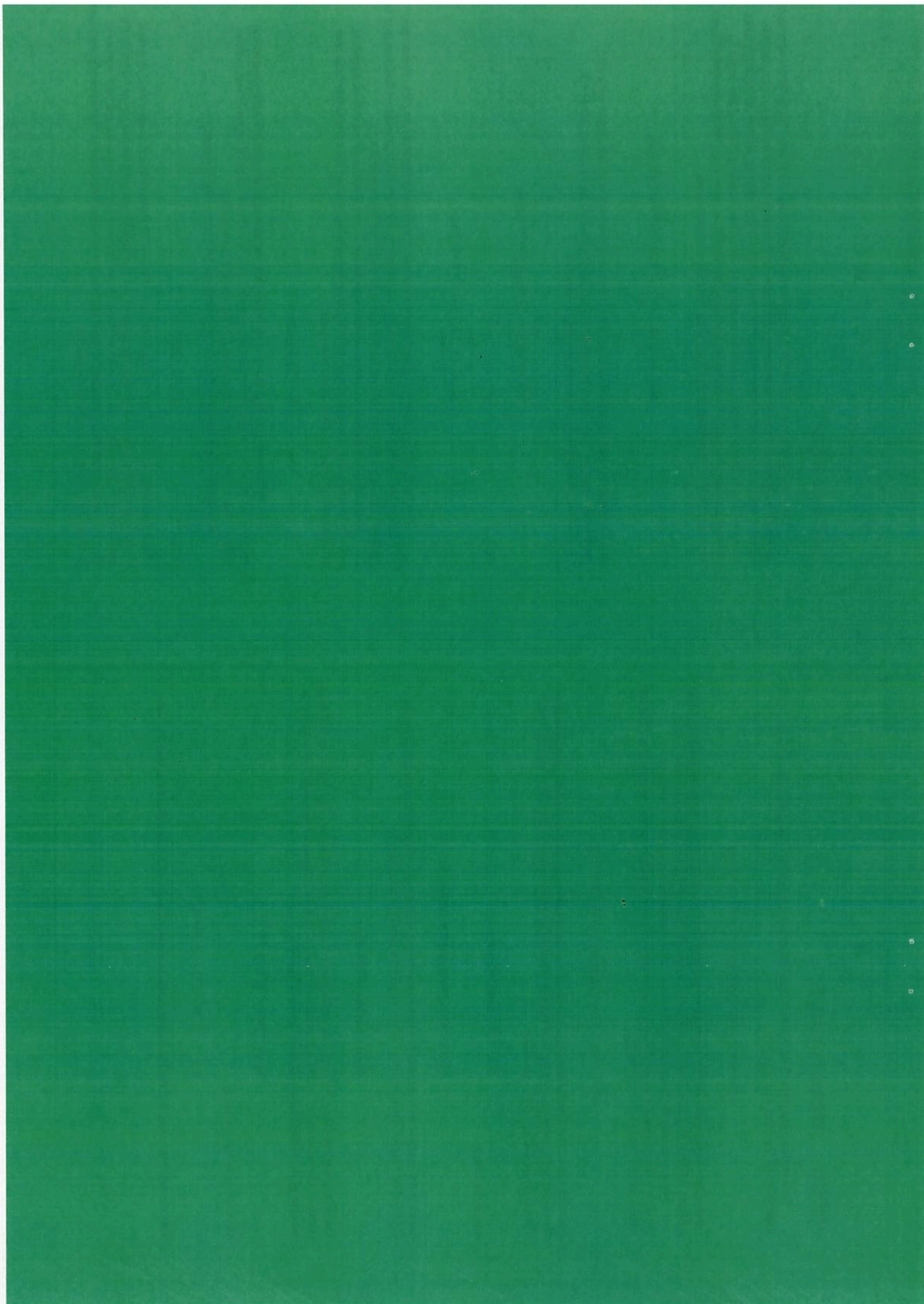
$$k_x = \int \gamma G_x dt = 0$$

Thus by sampling on both sides of the echo peak we automatically sample both positive and negative values of k_x . If the phase encoding gradient G_y is stepped between negative and positive values, both negative and positive values of k_y will also be sampled.

It is worth noting however that the equations for $S(k)$ possess the symmetry

$$S^*(-k) = S(k)$$

which follows as a result of the fact that $S(r)$ is real always. It follows that provided we quadrature detect to record both real and imaginary parts of the signal $S(k)$, we need record signals for only two quadrants in k space. Data for the other two can be constructed from them using the above relation. This is known as half NEX imaging because the number of signal acquisitions (excitations) and hence the total imaging time is effectively halved.



PNMR

Ref. 2

Theory

9. A Lock-In detection module can detect CW signals from solids with wide lines.
10. Both magnetic field and RF frequency sweeps are built in for studying CW resonance.
11. Digital clock stability in both the RF frequency and the pulse width synthesis are integral to the design.
12. The receiver recovery time (after pulse) has been improved by a factor of five to approximately 15 μ s, making signals from "soft" solids available for study.

The electronics system was designed by Dr. Norman Jarosik of the Princeton University Physics Department. Norman is a staff scientist in the "gravity Group" and the chief engineer of WMAP, the satellite that has been sensing and mapping the anisotropies in the microwave radiation left over from the Big Bang of the early universe. He also designed the PS-1 A and B whose well deserved reputation for reliability, as well as sensitivity, is known world-wide. Norman has been involved with TeachSpin from its 1992 inception.

Nuclear magnetic resonance is a large and mature field of physics. There certainly is a lot to learn and a real potential for a student to explore his or her own ideas in this area, perhaps even to find a real research project. We cannot think of a better instrument for students to pursue their own open-ended sets of experiments. But a new student should not be intimidated by the prospect of learning this vast subject. TeachSpin has outlined a manageable way to begin the study of NMR. Students can begin with straight-forward experiments that will develop their confidence and understanding and then branch out to more advanced measurements. This instrument is accessible, after all, it was designed from the moment of inception for the student. Enjoy!

B. OUTLINE OF THE PHYSICS

B.1 Brief History

In 1946, nuclear magnetic resonance (NMR) in condensed matter was discovered simultaneously by Edward Purcell at Harvard and Felix Bloch at Stanford using different instrumentation and techniques. Both groups, however, placed a sample containing magnetic nuclei in a uniform magnetic field and observed the response of those nuclei to a continuous (CW) radio frequency magnetic field as the field was tuned through resonance. This discovery opened up a new type of spectroscopy which has become one of the most important tools available to physicists, chemists, geologists, and biologists.

In 1950, Erwin Hahn, a postdoctoral fellow at the University of Illinois, again placing his sample of condensed matter in a uniform magnetic field, explored the response of the magnetic nuclei to pulse bursts of these same radio frequency (RF) magnetic fields. Hahn was interested in observing transient effects on the magnetic nuclei after the RF bursts. During these experiments, he observed a "spin echo" signal; that is, a signal from the magnetic nuclei that occurred after a two pulse sequence, at a time equal to the delay time between the two pulses. This discovery, and his brilliant analysis of the experiments, gave birth to a new technique for studying magnetic resonance. This pulse method originally had only a few practitioners, but now it is the method of choice for most laboratories. For the first twenty years after its discovery, continuous wave (CW) magnetic

resonance apparatus was used in almost every research chemistry laboratory, and no commercial pulsed NMR instruments were available. However, since 1966 when Ernst and Anderson showed that high resolution NMR spectroscopy can be achieved using Fourier transforms of the transient response, and cheap fast computers made this calculation practical, pulsed NMR has become the dominant commercial instrumentation for most research applications.

Widely used in physics and chemistry to characterize materials, NMR is a microscopic method in the sense that it probes the nuclei and their immediate surroundings. Within a certain solid, for instance, there may be a variety of local magnetic fields. A magnetic measurement by a magnetometer measures an average field. NMR, on the other hand, is capable of measuring the local field at atomic nuclei. Another example would be a molecule containing a several atoms of the same element, for instance carbon, in various different configurations. The NMR signal will be different for each one of these configurations. The microscopic nature of the NMR measurement makes it extremely useful, and often unique. Of course, in order to have a signal of detectable magnitude, we need many molecules of the same configuration to occur within the sample. The nuclei investigated in this set of experiments, hydrogen (proton) and fluorine, are very abundant and give particularly strong signals.

This technology has also found its way into medicine. MRI (magnetic resonance imaging; the word "nuclear" being removed to relieve the fears of the scientifically illiterate public) scans are revolutionizing radiology. This imaging technique seems to be completely noninvasive, produces remarkable three dimensional images, and has the potential to give physicians detailed information about the inner working of living systems. For example, preliminary work has already shown that blood flow patterns in both the brain and the heart can be studied without dangerous catheterization or the injection of radioactive isotopes. Someday, MRI scans may be able to pinpoint malignant tissue without biopsies. MRI is only in its adolescence, and we will see many more applications of this diagnostic tool in the coming years.

You have purchased a pulsed and CW NMR spectrometer designed specifically for teaching. The PS2-A is a complete spectrometer, including the magnet, magnet temperature controller, pulse generator, oscillator, pulse amplifier, sensitive receiver, linear amplitude and phase-sensitive detector, sample probe, gradient coils with dedicated current regulated supply and a CW lock-in detector. You need only supply the oscilloscope and the substances you wish to study. Now you are ready to learn the fundamentals of both CW and pulsed nuclear magnetic resonance spectroscopy.

Nuclear magnetic resonance is a vast subject. Tens of thousands of research papers and hundreds of books have been published on NMR. We will not attempt to explain or even to summarize this literature. Some of you may wish to do only a few simple experiments with the apparatus and achieve a basic conceptual understanding, while others may aim to understand the details of the density matrix formulation of relaxation processes and do some original research. The likelihood is that the majority of students will work somewhere in between these two extremes. In this section we will provide a brief theoretical introduction to many important ideas of PNMR. This will help you get started and can be referred to later. These remarks will be brief, not completely worked out from first principles, and not intended as a substitute for a careful study of the literature and published texts. An extensive annotated bibliography of important papers and books on the subject is provided at the end of this section.

B.2 Theory

Magnetic resonance is observed in systems where the magnetic constituents have **both a magnetic moment and an angular momentum**. Many, but not all, of the stable nuclei of ordinary matter have this property. In “classical physics” terms, magnetic nuclei act like a small spinning bar magnet. For this instrument, we will be concerned with only two nuclei, the nucleus of hydrogen, which is a single proton, and the nucleus of fluorine which contains both protons and neutrons. Both nuclei can be thought of as small spinning bar magnets with a magnetic moment μ and an angular momentum J , which are related by the vector equation:

$$\mu = \gamma J \quad (1.1)$$

The proportionality factor γ is called the “gyromagnetic ratio” and its values are unique to each kind of nucleus in the experiment.

The nuclear angular momentum is quantized in units of $\hbar$ as:

$$J = \hbar I \quad (1.2)$$

where I is the “spin” of the nucleus.

The magnetic energy U of the nucleus in an external magnetic field is:

$$U = -\mu \cdot B \quad (1.3)$$

If the magnetic field is in the z-direction, then the magnetic energy is:

$$U = -\mu_z B_0 = -\gamma \hbar I_z B_0 \quad (1.4)$$

Quantum mechanics requires that the allowed values I_z , m_I , be quantized as

$$m_I = I, I-1, I-2, I-3 \dots -I. \quad (1.5)$$

Both of the nuclei we are investigating, the proton (hydrogen nucleus) and the fluorine nucleus have spin one half ($I = 1/2$). Therefore, the allowed values of I_z are simply

$$m_I = \pm 1/2 \quad (1.6)$$

This means that there are only two magnetic energy states for these nuclei when residing in a constant magnetic field B_0 . These states are described in Figure 1.1.

The energy separation between the two states, ΔU , can be written in terms of an angular frequency or as

$$\Delta U = \hbar \omega_0 = \gamma \hbar B_0$$

or

$$\omega_0 = \gamma B_0 \quad (1.7)$$

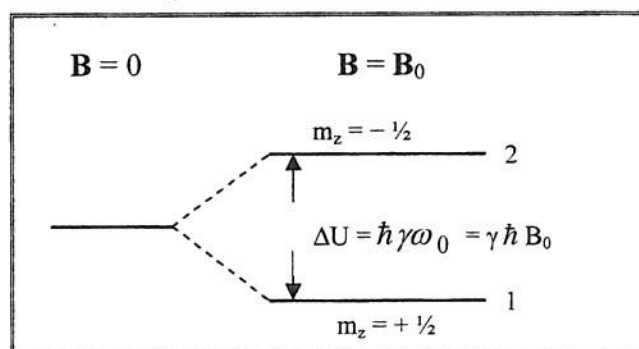


Figure 1.1 Energy State Separation in a Magnetic Field

Equation 1.7, $\omega_0 = \gamma B_0$ describes the fundamental resonance condition.

For the proton: $\gamma_{\text{proton}} = 2.675 \times 10^8 \text{ rad/sec-tesla}$ (1.8)

For Fluorine: $\gamma_{\text{fluorine}} = 2.517 \times 10^8 \text{ rad/sec-tesla}$

For the proton, the numerical relationship between the resonant frequency and the constant magnetic field is worth remembering:

$$f_{\text{proton}} (\text{MHz}) = 42.58 B_0 \text{ tesla} \quad (1.9)$$

If a one milliliter (ml) sample of water (containing about 7×10^{19} protons) is placed in a magnetic field in the z-direction, a nuclear magnetization in the z-direction eventually becomes established. This nuclear magnetization occurs because of unequal population of the two possible quantum states. If N_1 and N_2 are the number of spins per unit volume in the respective states, then the population ratio (N_2 / N_1), in thermal equilibrium, is given by the Boltzmann factor as:

$$\frac{N_2}{N_1} = e^{\frac{\Delta U}{kT}} = e^{\frac{\hbar \omega_0}{kT}} \quad (1.10)$$

and the magnetization is

$$M_z = (N_1 - N_2) \mu \quad (1.11)$$

The thermal equilibrium magnetization per unit volume for N magnetic moments is

$$M_0 = N\mu \tanh\left(\frac{\mu B}{kT}\right) \approx N \frac{\mu^2 B}{kT} \quad (1.12)$$

$$\text{where } N = N_1 + N_2$$

This magnetization does not appear instantaneously when the sample is placed in the magnetic field. It takes a finite time for the magnetization to build up to its equilibrium value along the direction of the magnetic field (which we define as the z-axis). For most systems, the z-component of the magnetization is observed to grow exponentially as depicted in Figure 1.2.

The differential equation that describes such a process assumes that the rate of approach to equilibrium is proportional to the difference between the equilibrium value M_0 and the instantaneous value $M_z(t)$:

$$\frac{dM_z(t)}{dt} = \frac{M_0 - M_z}{T_1} \quad (1.13)$$

where T_1 is called the spin-lattice relaxation time.

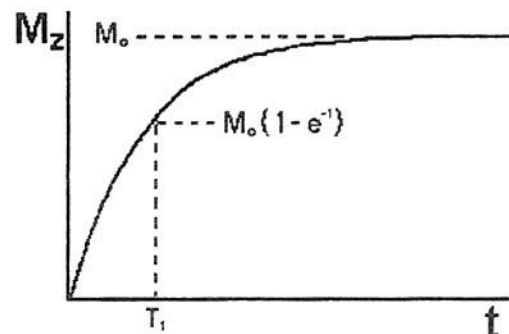


Figure 1.2 Magnetization vs. time for a sample placed in a magnetic field

If the unmagnetized sample is placed in a magnetic field, so that at $t = 0$, $M_z = 0$, then direct integration of equation 1.13, with these initial conditions, gives:

$$M_z(t) = M_0(1 - e^{-\frac{t}{T_1}}) \quad (1.14)$$

The rate at which the magnetization approaches its thermal equilibrium value is characteristic of the particular sample. Typical values range from microseconds to seconds. What makes one material take 10 microseconds to reach equilibrium while another material (also with protons as the nuclear magnets) takes 3 seconds? Obviously, some processes in the material make the protons “relax” towards equilibrium at different rates. The study of these processes is one of the major topics in magnetic resonance.

Although we will not attempt to discuss these processes in detail, a few ideas are worth noting. In thermal equilibrium, more protons are in the lower energy state than the upper. When the unmagnetized sample was first put in the magnet, the protons occupied the two states equally that is ($N_1 = N_2$). During the magnetization process energy must flow from the nuclei to the surroundings, since the magnetic energy from the spins is reduced. The surroundings which absorb this energy are referred to as “the lattice”, even for liquids or gases. Thus, the name “spin-lattice” relaxation time for the characteristic time of this energy flow.

However, there is more than energy flow that occurs in this process of magnetization. Each proton has angular momentum (as well as a magnetic moment) and the angular momentum must also be transferred from the spins to the surroundings during magnetization. In quantum mechanical terms, the “lattice” must have angular momentum states available when a spin goes from $m_1 = -1/2$ to $m_1 = +1/2$. In classical physics terms, the spins must experience a torque capable of changing their angular momentum. The existence of such states is usually the critical determining factor in explaining the enormous differences in T_1 for various materials. Pulsed NMR is ideally suited for making precise measurements of this important relaxation time. The pulse technique gives a direct and unambiguous measurement, where as CW spectrometers require a difficult, indirect, and imprecise technique to measure the same quantity.

What about magnetization in the x-y plane? In thermal equilibrium the only **net** magnetization of the sample is M_z , the magnetization along the external constant magnetic field. This can be understood from a simple classical model of the system. Think of placing a collection of tiny current loops in a magnetic field. The torque τ on the loop is $\mu \times B$ and that torque causes the angular momentum of the loop to change, as given by:

$$\tau = \frac{dJ}{dt} \quad \text{or} \quad \mu \times B = \frac{dJ}{dt} \quad (1.15)$$

For our nuclei, equation 1.15 becomes:

$$\boldsymbol{\mu} \times \mathbf{B} = \frac{1}{\gamma} \frac{d\boldsymbol{\mu}}{dt} \quad (1.16)$$

Equation 1.16 is the classical equation describing the time variation of the magnetic moment of the proton in a magnetic field. It can be shown from equation 1.16 that the magnetic moment will execute precessional motion, depicted in 1.3. The precessional frequency $\omega_0 = \gamma B_0$, is just the resonant frequency in equation 1.7.

If we add up all the magnetization for the 10^{20} nuclei in our sample in thermal equilibrium, the μ_z components sum to M_z , but the x and y components of the individual magnetic moments add to zero.

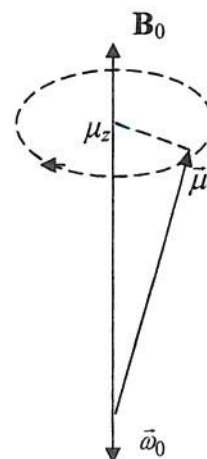


Figure 1.3

For the x-components of every nucleus to add up to some M_x , there must be a definite phase relationship among all the precessing spins. For example, we might start the precessional motion with the x-component of the spins lined up along the x-axis. But that is not the case for a sample simply placed in a magnet. In thermal equilibrium, the spin components in the x-y plane are oriented randomly. Thus, in thermal equilibrium there is no transverse (x or y) component of the net magnetization of the sample. As we shall soon see, however, there is a way to **create** such a transverse magnetization using radio frequency pulsed magnetic fields. The idea is to quickly rotate the thermal equilibrium magnetization M_z into the x-y plane and thus create a temporary M_x and M_y . Let's see how this is done.

Equation 1.16 can be generalized to describe the classical motion of the net magnetization of the entire sample. It then becomes

$$\frac{d\mathbf{M}}{dt} = \gamma \mathbf{M} \times \mathbf{B} \quad (1.17)$$

where $\mathbf{B}$ is any magnetic field, including time dependent rotating fields.

Suppose we apply not only a constant magnetic field $B_0 \hat{k}$, but a rotating (circularly polarized) magnetic field of angular frequency ω the x-y plane so the **total** field is written as [†]

$$\mathbf{B}(\mathbf{t}) = B_1 \cos \omega t \hat{i} + B_1 \sin \omega t \hat{j} + B_0 \hat{k} \quad (1.18)$$

The analysis of the magnetization in this complicated time dependent magnetic field can best be carried out in a non-inertial rotating coordinate system. The coordinate system of choice is rotating at the same angular frequency as the rotating magnetic field with its axis in the direction of the static magnetic field. In this **rotating coordinate system**, the rotating magnetic field appears to be stationary and aligned along the x^* axis (Figure 1.4). However, from the point of view of the rotating coordinate system, B_0 and B_1 are not the only magnetic fields. An **effective field** along the z^* direction, of $-\frac{\omega}{\gamma} \hat{k}^*$ must also be included.

Let's justify this new effective magnetic field with the following physical argument.

Equations 1.16 and 1.17 predict the precessional motion of a magnetization in a constant magnetic field $B_0 \hat{k}$. Suppose one observes this precessional motion from a rotating coordinate system which rotates **at the precessional frequency**. In this frame of reference the magnetization appears stationary, in some fixed position. The only way a magnetization can remain fixed in space is if there is no torque on it. If the magnetic field is zero in the reference frame, then the torque on $\mathbf{M}$ is always zero no matter what direction $\mathbf{M}$ is oriented. The magnetic field is zero (in the rotating frame) if we add the effective field $-\frac{\omega}{\gamma} \hat{k}^*$

which is equal to $B_0 \hat{k}^*$

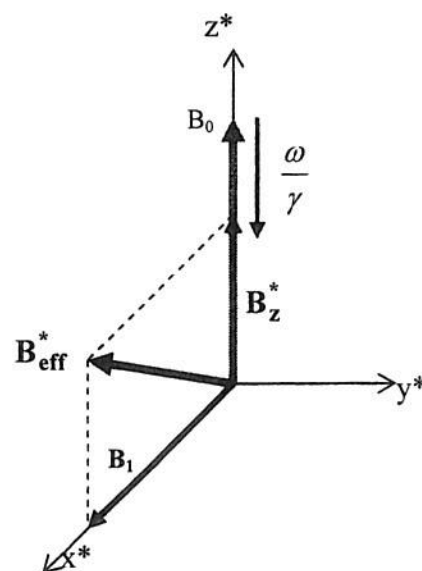


Figure 1.4

Transforming the magnetic field expression in equation (1.18) into such a rotating coordinate system, the total magnet field in the rotating frame $\mathbf{B}^*$ is

$$\mathbf{B}^*_{\text{eff}} = B_1 \hat{i}^* + (B_0 - \frac{\omega}{\gamma}) \hat{k}^* \quad (1.19)$$

Figure 1.4 is a representation of Equation 1.19. The classical equation of motion of the magnetization as observed in the rotating frame is then

$$\left. \frac{d\mathbf{M}}{dt} \right|_{\text{rot}} = \gamma \mathbf{M} \times \mathbf{B}^*_{\text{eff}} \quad (1.20)$$

which shows that $\mathbf{M}$ will precess about $\mathbf{B}^*_{\text{eff}}$ in the rotating frame.

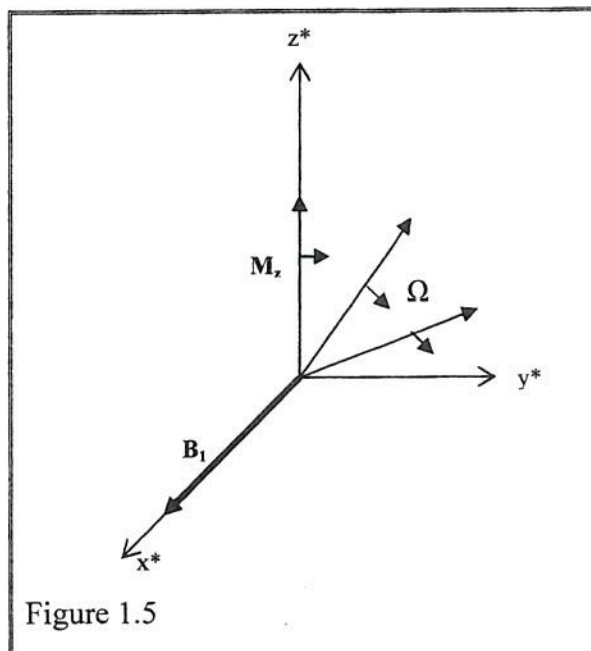
[†]What is actually applied is an oscillating field $2B_1 \cos \omega t \hat{i}$. But that can be decomposed into two counter rotating fields $B_1 (\cos \omega t \hat{i} + \sin \omega t \hat{j}) + B_1 (\cos \omega t \hat{i} - \sin \omega t \hat{j})$. One of the counter rotating fields can be shown to have no practical affects on the spin system and can be ignored in this analysis.

Suppose now, we create a rotating magnetic field at a frequency ω_0 as such that

$$\frac{\omega}{\gamma} = B_0 \quad \text{or} \quad \omega = \gamma B_0 = \omega_0 \quad (1.21)$$

In that case, $\mathbf{B}^*_{\text{eff}} = B_1 \hat{i}^*$, a constant magnetic field in the x^* direction (Figure 1.5). Then, the magnetization M_z begins to precess about this magnetic field at a rate $\Omega = \gamma B_1$, (in the rotating frame). If we **turn off** the B_1 field at the instant the magnetization reaches the x-y plane, we will have created a transient (non-thermal equilibrium) situation where there **is** a net magnetization in the x-y plane. If this rotating field is applied for twice the time the transient magnetization will be $-M_z$ and if it is left on four times as long the magnetization will be back where it started, with M_z along the z^* axis. These are called:

$$\begin{aligned} 90^\circ \text{ or } \pi/2 \text{ pulse} &\rightarrow M_z \rightarrow M_y \\ 180^\circ \text{ or } \pi \text{ pulse} &\rightarrow M_z \rightarrow -M_z \\ 360^\circ \text{ or } 2\pi \text{ pulse} &\rightarrow M_z \rightarrow M_z \end{aligned}$$



In the laboratory (or rest) frame, where the experiment is actually carried out, the magnetization not only precesses about B_1 , but rotates about $\hat{k}$ during the pulse. **It is not possible, however, to observe the magnetization during the pulse.** Pulsed NMR signals are observed **AFTER** the transmitter pulse is over? But, what is there to observe **AFTER** the transmitter pulse is over? The spectrometer detects the **net magnetization precessing about the constant magnetic field $B_0 \hat{k}$ in the x-y plane. Nothing Else!**

Suppose a 90° ($\pi/2$) pulse is imposed on a sample in **thermal equilibrium**. The net equilibrium magnetization will be rotated into the x-y plane where, after the pulse, it will precess about $B_0 \hat{k}$. But the x-y magnetization will not last forever. For most systems, this magnetization decays exponentially as shown in Figure 1.6. The differential equations which describe the decay **in the rotating coordinate system** are:

$$\frac{dM_{x^*}}{dt} = -\frac{M_{x^*}}{T_2} \quad \text{and} \quad \frac{dM_{y^*}}{dt} = -\frac{M_{y^*}}{T_2} \quad (1.22)$$

whose solutions are

$$M_{x^*} = M_0 e^{-\frac{t}{T_2}} \quad \text{and} \quad M_{y^*} = M_0 e^{-\frac{t}{T_2}} \quad (1.23)$$

where the characteristic decay time T_2 is called the **Spin-Spin Relaxation Time**.

One simple way to understand this relaxation process, from the classical perspective, is to recall that each nucleus is itself a magnet and produces a magnetic field at its neighbors. Therefore, for a given distribution of nuclei, there must also be a distribution of *local* fields at the various nucleus sites. Thus, the nuclei precess about $B_0 \hat{k}$ with a distribution of frequencies, not a single frequency ω_0 .

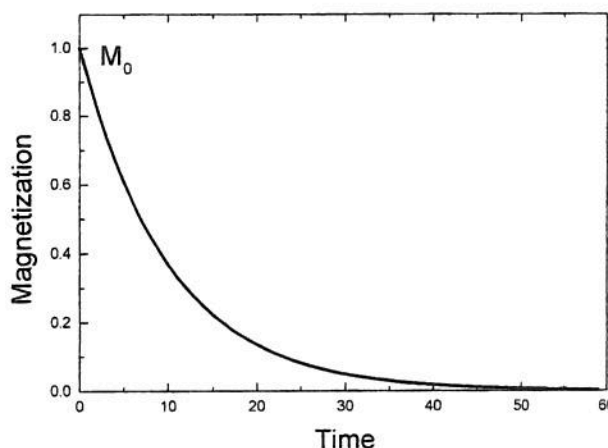


Figure 1.6

Even if all the nuclei begin in phase (after the 90° pulse), they will soon get out of phase and the net x-y magnetization will eventually go to zero. A measurement of T_2 , the decay constant of the x-y magnetization, gives information about the distribution of local fields at the nuclear sites.

From this analysis, it would appear that the spin-spin relaxation time T_2 can be determined by simply plotting the decay of M_x (or M_y) after a 90° pulse. This signal is called the *free precession or free induction decay* (FID). If the field of the magnet were perfectly uniform over the entire sample volume, then the time constant associated with the free induction decay would, in fact, be T_2 . But, in most cases, it is the nonuniformity of the magnet's field over the sample that is responsible for the observed decay constant of the FID. At its center, the PS2-A's magnet has sufficient uniformity to produce at least a .3 millisecond decay time. Using the electric shim coils (See Section II.F), students can improve the homogeneity so that decay time due to the magnet (called T_2^* in the jargon) is as long as 5 milliseconds, and, possibly, longer. Thus, for a sample whose $T_2 < 5$ ms, the free induction decay constant is also the T_2 of the sample. But what if T_2 is actually 5 msec or longer? The observed decay will still be about 5 ms. Here is where the genius of Erwin Hahn's discovery of the spin echo plays its crucial role.

Before the invention of pulsed NMR, the only ways to measure the real T_2 were to improve the magnet's homogeneity and to make the sample smaller. But, PNMR changed this.

Suppose we use a two pulse sequence, the first one 90° and the second one, turned on a time τ later, a 180° pulse. What happens? Figure 1.7 shows the pulse sequence and Figure 1.8 shows the progression of the magnetization in the rotating frame.

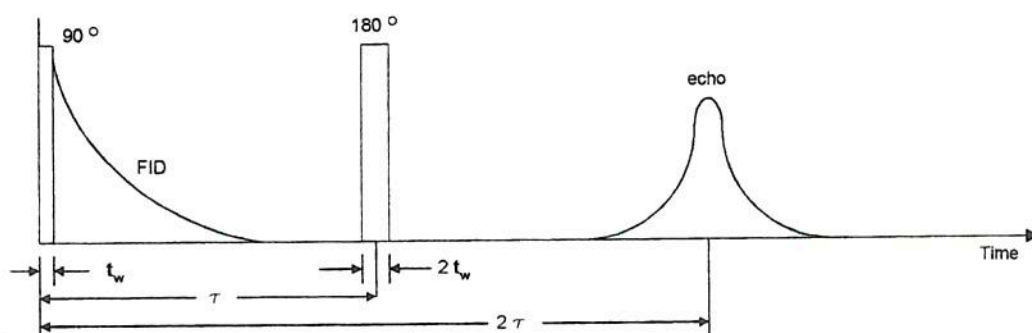


Figure 1.7: A 90° - 180° Pulse Sequence

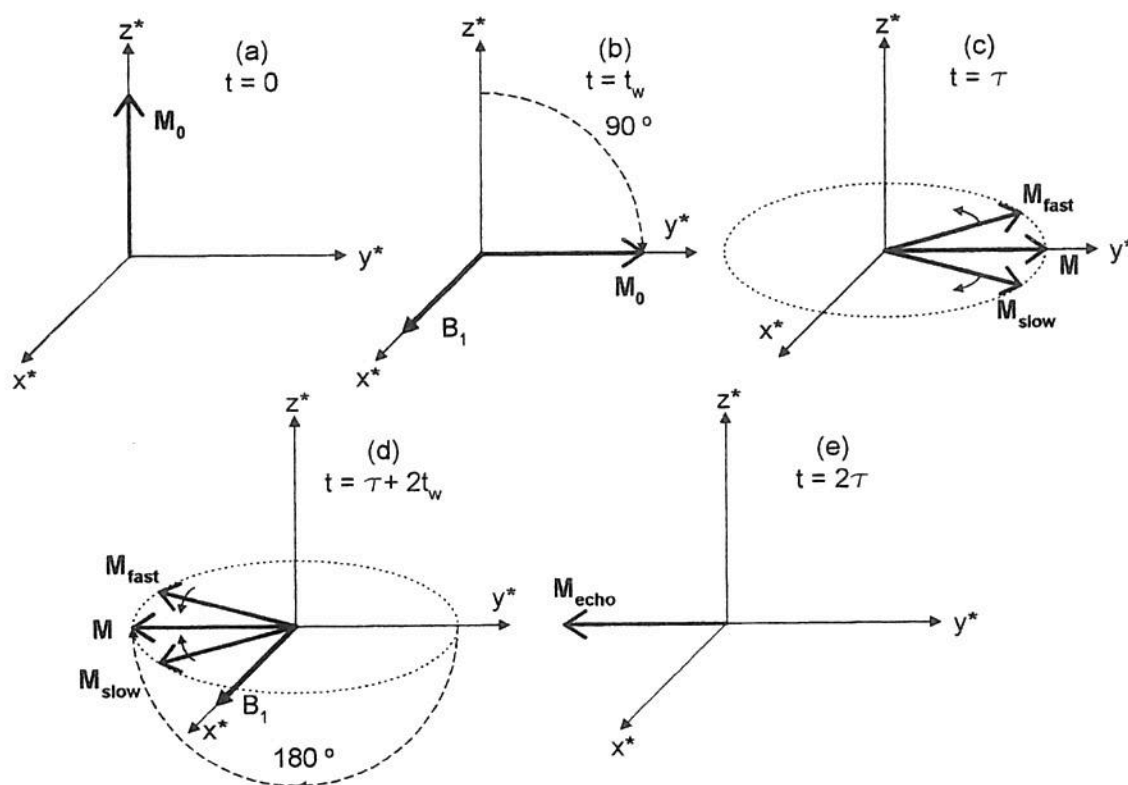


Figure 1.8: Progression of the Magnetization in the Rotating Frame

- Thermal equilibrium magnetization M_0 along the z axis before the rf pulse.
- M_0 rotated to the y -axis after the 90° pulse.
- The magnetization in the x - y plane is decreasing because some of the spins Δm_{fast} are in a higher field, and other spins Δm_{slow} are in a lower field static field.
- The spins are rotated 180° (visualize flipping the entire x - y plane like a pancake on the griddle) by the pulsed rf magnetic field.
- The rephasing the three magnetization "bundles" to form an *echo* at $t = 2t$.

Study the diagrams in Figures 1.7 and 1.8 carefully. The 180° pulse allows the x-y magnetization to rephase to the value it would have had with a perfect magnet.

The echo process is analogous to an egalitarian and PC foot race for a kindergarten class, a race that makes all the children in the class winners, no matter how fast they can run. What if the race had the following rules? All the children are to line up at the starting line. At the first whistle, they are to run as fast as they can down the field. At the second whistle, they are to turn around and run back toward the starting line. First person back wins!! Of course, it is a tie, except for the ones who “interfere” with one another or fall down. As the children run away, the field spreads out with the fastest ones getting farther and farther ahead. At some point there is no semblance of order. On the trip back, as the faster ones overtake the slow pokes, who are now in the lead, the group comes together again “rephasing” as they pass the start line.

In Pulsed NMR, the 180° pulse is like that whistle. The spins in areas of larger field get out of phase by $+\Delta\theta$ in a time τ . After the 180° pulse, they continue to precess faster than M while the slower precessing spins do just the opposite. At 2τ all the spins return to the in-phase condition and then again dephase.

Yet some loss of M_{xy} magnetization has occurred and the maximum height of the echo is not the same as the maximum height of the FID. This loss of transverse magnetization occurs because of stochastic fluctuation in the local fields at the nuclear sites which is not rephasable by the 180° pulse. These are the real T_2 processes that we are interested in measuring. To find the “real” T_2 , we use a series of $90^\circ - \tau - 180^\circ$ pulse experiments, varying τ , and then plotting the echo height as a function of time between the FID and the echo.

The transverse magnetization as measured by the maximum echo height is written as:

$$M_{xy}(2\tau) = M_0 e^{-\frac{2\tau}{T_2}} \quad (1.24)$$

That’s enough theory for now. **Let’s summarize:**

1. Magnetic resonance is observed in systems whose constituent particles have **both** a magnetic moment and angular momentum.
2. The resonant frequency of the system depends on the applied magnetic field in accordance with the relationship $\omega_0 = \gamma B_0$ where

$$\begin{aligned} \gamma_{\text{proton}} &= 2.675 \times 10^8 \text{ radian/sec-tesla} \\ \text{or} \\ f_{\text{proton}} &= 42.58 \text{ MHz/tesla} \quad (\text{for protons}) \\ f_{\text{fluorine}} &= 40.055 \text{ MHz/tesla} \quad (\text{for fluorine}) \end{aligned}$$

3. The thermal equilibrium magnetization is parallel to the applied magnetic field, and approaches equilibrium following an exponential rise characterized by the constant T_1 the spin-lattice relaxation time.

4. Classically, the magnetization (which is the vector sum of the individual magnetic moments of the nuclei) obeys the differential equation

$$\frac{d\mathbf{M}}{dt} = \gamma(\mathbf{M} \times \mathbf{B})$$

where $\mathbf{B}$ may be a time dependent field.

5. Pulsed NMR employs a rotating radio frequency magnetic field described by

$$\mathbf{B}(t) = B_1 \cos \omega t \hat{i} + B_1 \sin \omega t \hat{j} + B_0 \hat{k}$$

6. The easiest way to analyze the motion of the magnetization during and after the rf pulsed magnetic field is to transform into a rotating coordinate system. If the system is rotating at an angular frequency ω along the direction of the magnetic field, a fictitious magnetic field must be added to the real fields such that the total effective magnetic field in the rotating frame is:

$$\mathbf{B}_{\text{eff}}^* = B_1 \hat{i}^* + (B_0 - \frac{\omega}{\gamma}) \hat{k}^*$$

7. On resonance $\omega = \omega_0 = \gamma B_0$ and $\mathbf{B}_{\text{eff}}^* = B_1 \hat{i}^*$. In the rotating frame, during the pulse, the spins precess around $\mathbf{B}_1^*$.
8. A 90° pulse is one where the pulse is left on just long enough (t_w) for the equilibrium magnetization M_0 to rotate to the x-y plane. That is;

$$\omega_1 t_w = \pi/2 \text{ radians or } t_w = \frac{\pi}{2\omega_1}$$

But

$$\omega_1 = \gamma B_1 \quad (\text{since, on resonance, } B_1 \text{ is the only field in the rotating frame.})$$

So,

$$t_w(90^\circ) = \frac{\pi}{2\gamma B_1} \quad \text{duration of the } 90^\circ \text{ pulse} \quad (1.25)$$

9. T_2 - the spin-spin relaxation time - is the characteristic decay time for the nuclear magnetization in the x-y (or transverse) plane.
10. The spin-echo experiments allow the measurement of T_2 in the presence of a nonuniform static magnetic field. For those cases where the free induction decay time constant, (sometimes written T_2^*) is shorter than the real T_2 , the decay of the echo envelope's maximum heights for various times τ , gives the real T_2 .

C. REFERENCES

The following is a rather long list, and you certainly will not have the time all of them. In fact, you may only have time to read a rather small percentage of what is listed. But, you must take the time to read some of them so you have a basic understanding of magnetic resonance spectroscopy.

C.1 Books

C.P. Slichter: "Principles of Magnetic Resonance" Springer Series in Solid-State Sciences 1
Third Edition (1990) Springer-Verlag

A complete text with problems, clear explanations, appropriate for advanced undergraduate or graduate level students. **Any serious student of magnetic resonance should own it.** Everyone should read at least some of it. This reference contains a nearly complete bibliography of the important papers published in both NR and ESR spectroscopy. Consult this text for references to particular subjects.

T.C. Farrar, E.D. Becker: "Pulsed And Fourier Transform NMR", Academic Press 1971
A good introduction, with simplified mathematics, to the subject. Gives students a physical feel for the basic ideas of PNMR.

G. E. Pake and T. L. Estle: "The Physical Principles of Electron Paramagnetic Resonance",
Benjamin-Cummings, Menlo Park CA (1978)
Don't let the title ESR scare you away from using this excellent text. It has clear discussions of important ideas of magnetic resonance, such as the rotating coordinate systems etc.

R. T. Schumacher: "Introduction to Magnetic Resonance", Benjamin-Cummings,
Menlo Park CA 1970.

N. Bloembergen: "Nuclear. Magnetic Relaxation", W.A. Benjamin, New York 1961
This is Bloembergen's Ph.D. thesis, reprinted, but it is like no other thesis you will ever read. Describes some of the classic ideas of magnetic resonance, still very worth reading, you will see why he is a Nobel Laureate.

A. Abragam: "Principles of Nuclear Magnetism", Clarendon, Oxford 1961
This text is in a class by itself, but not easy for the beginner. Abragam has his own way of describing NMR. Important, but clearly for advanced students.

E. R. Andrew, "Nuclear Magnetic Resonance &' Cambridge University Press, New York, 1956
A good general discussion of theory, experimental methods, and applications of NMR.